# How LLMs Feel About Applicants: Capturing Social Intuition as Vectors

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**Abstract**

Recent interpretability work shows that emotions can be represented as linear directions in a language model's residual stream. We adapt this approach to social perception across nine open-weight models from the Gemma, Qwen, and Llama families. Using synthetic stories, we derive warmth and competence directions and then test whether name-level projections align with human ratings and whether steering these directions changes callback recommendations. Human alignment was model-dependent: both dimensions aligned positively in Gemma-3 and Qwen3.6, whereas Llama-3.1 and Qwen3-14B inverted warmth, and Gemma-4 showed mixed alignment across checkpoints. Hiring steering was similarly heterogeneous: positive interventions increased callback margins in some checkpoints, decreased them in others, and produced nonlinear or decision-stable responses elsewhere. Thus, a common extraction procedure recovers warmth and competence directions across models, but their correspondence with human perceptions and causal influence on hiring decisions are not uniform.

## Introduction

*Work in progress.*

## Background: Reading a Concept Out of a Language Model

**From Layers to Directions.** A large language model processes a piece of text by passing it through a long stack of layers, and each layer reads from and writes back to a shared running representation known as the residual stream (Vaswani et al., 2017; Elhage et al., 2021). Think of the residual stream as a notebook passed from one layer to the next: every layer reads what earlier layers have written, adds its own notes, and hands the notebook onward. At any chosen depth, this running state is simply a long list of numbers, a high-dimensional vector, and circuit-level analyses have shown that specific features of the input, from low-level syntax to high-level semantic content, leave consistent geometric traces in that vector across many different contexts (Olah et al., 2020). The idea that such traces take the form of *directions* rather than isolated coordinates goes back to word-embedding analogies, where arithmetic on word vectors captures semantic relationships (Mikolov et al., 2013), and the *linear representation hypothesis* extends this claim to modern transformer models: a high-level feature corresponds to a direction in the residual stream, and projecting the model's state onto that direction yields a scalar measuring how strongly the feature is present (Park et al., 2024). Extracting a direction for some target concept is then a matter of contrast: we collect residual-stream states from texts that express the concept (condition $A$) and from matched texts that do not (condition $B$), and take the mean difference between the two groups,

$$v = \bar{h}_A - \bar{h}_B. \tag{1}$$

Normalizing $v$ to a unit vector $\hat{v}$ gives a concept direction, and the concept score of any new piece of text is the projection $\hat{v}^\top h$. The claim worth testing, and the one we validate here through cross-validated classification and topic-holdout accuracy, is that a direction built this way generalizes beyond the handful of texts used to construct it.

**Emotion Vectors.** A direction built purely from the mean-difference of Eq. 1 only tells us that high- and low-condition texts sit in different places; it does not yet tell us whether the model actually relies on that direction when it acts. Testing that requires an intervention rather than an observation: at a chosen layer, we can override the model's own internal state during inference,

$$h' = h + \alpha\,\hat{v}, \tag{2}$$

where $\alpha$ is a signed scalar that sets how strongly, and in which direction, we push. If the model's downstream behavior then shifts systematically as $\alpha$ changes, while an unrelated, orthogonal direction produces no comparable shift, the direction is not merely correlated with the concept but participates causally in the model's computation. Activation addition (Turner et al., 2023) and representation engineering (Zou et al., 2023) established this intervene-and-observe recipe and

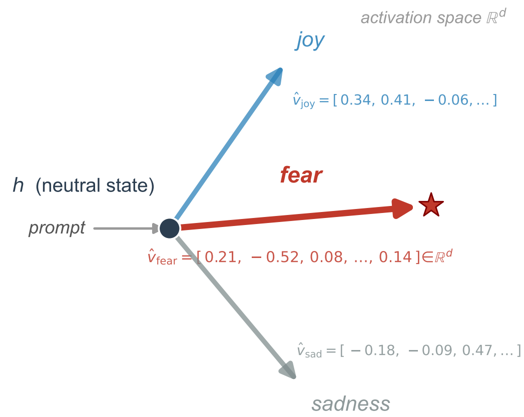


**Figure 1: Emotion vectors as directions in activation space (schematic).** Conceptually, an emotion vector starts from a neutral model state $h$. A prompt that evokes a given emotion, such as fear, pushes the residual stream toward a direction $\hat{v}_{\text{fear}}$, encoded here with its representative $d$-dimensional coordinates. This direction behaves like a commuter who takes the same route to work every day: the attitude, direction, and flow stay the same each time. Across thousands of different prompts that express the same emotion in different words, the model converges on this same direction through its layers, and it produces its output by moving along that direction. After Sofroniew et al. (2026), who call this an emotion vector; we reuse the geometry, not the construct.

demonstrated it across a range of models and behaviors. Most recently, Sofroniew et al. (2026) applied exactly this recipe to emotion concepts in Claude Sonnet: building contrastive text pairs for each emotion, they extracted mean-difference directions in the residual stream, pushed the model's state along those directions during inference, and observed systematic, interpretable shifts in the model's behavior. Their study also introduced a neutral-corpus PCA denoising procedure that strips out general evaluative valence from a direction vector, a step we adopt to separate warmth and competence content from shared narrative polarity (detailed in the Supplementary Materials). Fig. 1 illustrates the resulting picture: each emotion occupies its own direction in residual-stream space, with a numeric representation that can be extracted, inspected, and used to steer the model.

**From Emotions to Hiring.** Nothing in this read-and-steer recipe is specific to emotion. Any construct that is linearly encoded in the residual stream should, in principle, be extractable by the same mean-difference procedure and testable by the same steering intervention, whether that construct is an emotional state or a dimension of social perception. Hiring is a domain where such a construct would matter immediately: behavioral audits have already documented that large language models produce uneven callback recommendations across demographic groups (An et al., 2024; Gaebler et al., 2024), yet none of that work locates the representation inside the model that might be doing the work. We therefore ask whether a social-perception construct known to predict human callback outcomes is encoded the same way emotion concepts are, and whether steering it changes a model's own callback recommendations. The next section describes how we chose that construct and built the materials needed to test it.

# Adapting the Method to Hiring

**Why Warmth and Competence.** The Stereotype Content Model proposes that social perception organizes along two fundamental axes: warmth, capturing perceived intentions and trustworthiness, and competence, capturing perceived ability and effectiveness (Fiske et al., 2002). Evidence across cultural contexts suggests that warmth and competence are recurring dimensions of both interpersonal impressions and perceptions of social groups (Fiske et al., 2007; Cuddy et al., 2008). Warmth and competence therefore give us two well-established, independently motivated axes to search for inside a language model, rather than an ad hoc pair invented for this study.

**The Link to Hiring.** Gallo et al. (2024) provide the empirical bridge that connects these two axes to real labor-market outcomes. Their released ratings data contain 16,557 warmth and competence judgments from 787 raters covering 282 applicant first names drawn from nine source studies. For the eight name-based correspondence studies with usable callback data, a more favorable first principal component combining warmth and competence was moderately associated with higher callback rates on average. The effect nevertheless varied substantially across studies, and its prediction interval included negative relationships. These findings provide a direct, quantitative connection between the social-perception dimensions we probe and measured labor-market outcomes, while showing that the relationship is not uniform. Behavioral audits of LLMs have separately documented name-driven callback disparities (An et al., 2024; An et al., 2025; Gaebler et al., 2024). To our knowledge, these studies have not tested whether internal warmth and competence representations carry that signal. We use the dataset released by Gallo et al. in two ways: as an external validity check on our probes, and as the reference point against which model-level disparities are compared.

**Concept Stories Without Identity.** Probing warmth and competence directly from hiring prompts would confound the social-perception dimensions with the hiring decision itself, so the direction vectors have to come from a separate set of non-hiring materials. Identity control is essential here. In unconstrained pilot generations, the language model systematically assigned male, Anglo-named protagonists to low-warmth and low-competence stories, which would have introduced demographic expectations before the hiring evaluation. We therefore use nameless, gender-neutral protagonists so that the vectors reflect depicted behavior rather than the demographic signal tested later.

**Applicant Names.** The hiring evaluation itself draws on a separate stimulus set: the 282 applicant first names from Gallo et al. (2024), each inserted into an otherwise identical, fixed job application so that the name is the only signal that varies across trials. Of these 282 names, 149 also appear in the published per-name callback dataset from the same source and carry US race and gender labels; this subset forms the benchmark comparison for demographic disparity. The full set of 282 is used for the probe-versus-human alignment check, since every name carries crowdsourced warmth and competence ratings even when published callback data are unavailable.

# Methods

**Story Preparation.** The corpus contains 200 narratives of approximately 90–150 words, divided evenly among high warmth, low warmth, high competence, and low competence (50 stories per condition). Its 50 everyday topics span ten domains, including workplace, learning, community, sport, and travel; each topic appears once in every condition to balance situational vocabulary across poles (Tab. S.1 lists every topic with its word-count range). Following the identity controls described above, every protagonist is nameless and referred to only as "they." A large language model generated the stories, which were then checked automatically for forbidden vocabulary, length balance, and demographic neutrality. Tab. S.2 records the generation and system prompt supplied to that model. For a concrete sense of the corpus, Tab. S.3 presents one story from each of the four conditions on a single shared topic, with a link to the complete set of 200 texts.

**Building the Warmth and Competence Vectors.** We extracted residual-stream activations from nine open-weights instruction-tuned models spanning three model families and two version generations each for Gemma and Qwen: Gemma-3-12B-it and Gemma-3-27B-it (Gemma Team, 2025), Gemma-4-12B, Gemma-4-26B-A4B, and Gemma-4-31B; Qwen3-14B (Yang et al., 2025), Qwen3.6-27B, and Qwen3.6-35B-A3B; and Llama-3.1-8B-Instruct (Grattafiori et al., 2024). Activations were read with TransformerLens (Nanda and Bloom, 2022) for the Gemma-3, Qwen3-14B, and Llama-3.1 checkpoints, with its Bridge interface for the three Gemma-4 checkpoints, and with a native Hugging Face backend for the two Qwen3.6 checkpoints, which the released TransformerLens build does not yet support. The *Backend* column of Tab. 1 lists the tool used for each checkpoint. Sofroniew et al. (2026), whose emotion-vector method motivates our approach, describe a progression from token-level emotional connotations in early layers to context-integrated representations in middle-to-late layers. They report most analyses at a layer approximately two thirds through the model, where their evidence indicates that emotion representations encode abstract content relevant to upcoming token predictions. Guided by this result, we operationalized the same relative depth as probe_layer_frac = 0.66 and fixed it across models before conducting the layer sweeps, rather than selecting a model-specific optimum. Integer rounding placed the probe between 63 and 66 percent of model depth; Tab. 1 reports the exact layer index, model width, and mean residual-stream norm for each model. For each story, activations were mean-pooled across token positions from position 50 to the end of the story, giving one $d_{\text{model}}$-dimensional vector per story. This offset was fixed in advance and applied uniformly across models rather than tuned per architecture: the earliest token positions carry input-agnostic outlier activations that would otherwise dominate the pooled mean (Sun et al., 2024; Xiao et al., 2024), and because the concept stories ran roughly 100 to 170 tokens, starting at position 50 discards about the first third of each narrative so that the pooled state reflects text where the warmth or competence content is already established. High- and low-condition stories cluster in distinct regions of this space (Fig. 2). The warmth and competence direction vectors are simply the mean difference between the two clusters:

$$v_W = \bar{x}_{\text{high warmth}} - \bar{x}_{\text{low warmth}},$$
$$v_C = \bar{x}_{\text{high comp.}} - \bar{x}_{\text{low comp.}}$$

where $\bar{x}_{\text{high warmth}}$ is the average pooled activation across the 50 high-warmth stories, $\bar{x}_{\text{low warmth}}$ is the same average across the 50 low-warmth stories, and $v_W$ (respectively $v_C$ for competence) is the resulting direction vector. The high and low conditions are a requirement of this construction rather than an incidental design choice: a mean-difference direction can only be recovered from a contrast, so each concept needs an opposing pair of story sets to define which way its vector points. A single graded warmth or competence scale would leave nothing to subtract. All nine models used the same stories, pooling rule, and prespecified seed (20260527), enabling comparison across architectures.

We check that each direction is a genuine, generaliz-

| Model | $d_{\text{model}}$ | Probe layer | $\|\bar{h}\|$ | Backend |
|---|---|---|---|---|
| Gemma-3-12B-it | 3840 | 31/48 | 79,722 | TL |
| Gemma-3-27B-it | 5376 | 40/62 | 61,576 | TL |
| Gemma-4-12B | 3840 | 31/48 | 97.8 | Bridge |
| Gemma-4-26B-A4B | 2816 | 19/30 | 68.4 | Bridge |
| Gemma-4-31B | 5376 | 39/60 | 188.5 | Bridge |
| Qwen3-14B | 5120 | 26/40 | 206.6 | TL |
| Qwen3.6-27B | 5120 | 42/64 | 71.9 | HF |
| Qwen3.6-35B-A3B | 2048 | 26/40 | 3.6 | HF |
| Llama-3.1-8B-Instruct | 4096 | 20/32 | 11.4 | TL |

**Table 1:** Model architecture and activation scale for nine checkpoints. $d_{\text{model}}$ is the number of activation values carried at each token, or the representation width. Probe layer gives the zero-indexed block used to build the directions and the total blocks; 31/48 means index 31 of a 48-block model. $\|\bar{h}\|$ is the typical activation size there: for each of 200 stories, token activations are averaged and the resulting vector's length is measured; the table reports the mean of these lengths. This value calibrates steering to each model's own scale, so larger values do not imply better or stronger models. Backend is the tool used to read activations: TL is TransformerLens, Bridge is its Hugging Face Bridge interface, and HF a native Hugging Face backend used for the Qwen3.6 checkpoints that TransformerLens does not yet support.

able signal rather than an artifact of the specific stories used to build it, using five complementary checks:

- **5-fold cross-validated accuracy**: We need to know whether the direction we found is picking up on something real, or whether it just memorized the exact 200 stories we happened to write. The 100 warmth stories (50 high, 50 low) are split into 5 equal groups of 20. In each round, a classifier is trained on 4 of the groups and tested on the 5th group, which it has never seen; this repeats 5 times so every group is tested once. Guessing randomly would get the label right about 50% of the time, so a classifier that gets far more than half of these unseen stories right is reading a real high-versus-low pattern in the stories, not reciting stories it has already seen.
- **Topic-holdout accuracy**: The check above still lets a classifier study some stories about a topic, say a stressful team meeting, before being tested on other stories about that same topic; getting those right could mean it recognized the topic, not the tone. This check closes that gap. The 50 topics are grouped so that every story about a topic, high- and low-condition alike, is held out together, and each test round contains topics the classifier has never read a single sentence about, in either condition. A classifier that still gets these completely unfamiliar topics right is recognizing warmth or competence itself, not the leftover vocabulary of a topic it has already studied.
- **Cohen's $d$**: The two checks above only tell us whether high- and low-condition stories can be told apart, not how far apart they really are. Each story is projected onto the direction, reducing it to a single number, and Cohen's $d$ measures the gap between the high- and low-condition averages on that number line. To know whether that gap is meaningful, we generate 1,000 random directions instead of the real one and compute the same gap for each. Every one of those 1,000 random directions produces a much smaller gap than our real direction does, which means the separation is not a coincidence.
- **Split-half cosine stability**: A direction built from one arbitrary set of stories should not depend on exactly which stories happened to be written. Each condition's 50 stories are randomly split into two non-overlapping halves of 25, and a separate direction is built from each half using only that half's stories. The cosine between these two independently-built directions tells us how similar they are. Across all nine models, this cosine is consistently high (roughly 0.69 to 0.82 for warmth, 0.83 to 0.90 for competence), meaning the direction reflects a stable property of the concept rather than an artifact of which 25 stories happened to end up in each half.
- **Cross-axis classification**: If warmth and competence were fully independent, knowing a story's position along the warmth direction would tell us nothing about whether it is high- or low-competence. To test this, competence stories are projected onto the warmth direction, and that single number is used to guess the competence label; the same test is then run in the other direction. Across all nine models this guess is highly accurate, typically 0.82 to 1.00, which means the two directions are not fully independent: they share a substantial amount of evaluative content rather than capturing two fully separate constructs.

Tab. 2 reports Cohen's $d$, the random-null $z$-score, split-half cosine, and calibrated cross-axis accuracy for all nine models; 5-fold cross-validated and topic-holdout accuracy reach 1.00 for every model on both axes and are reported in Results rather than repeated in the table. Two further analyses look beyond a single model's own directions:

- **Cross-model Spearman agreement**: Nine models, trained independently by different organizations, could each encode warmth as their own unrelated pattern, with no reason for their story rankings to agree with one another. To test this, every pair of the nine models is compared: each story is projected onto both models' own directions to give one score per model, and the two resulting sets of scores are correlated with Spearman's rank correlation. All thirty-six pairs show positive agreement on both axes, which means the nine architectures are converging on a shared cross-architecture construct rather than nine independent, unrelated ones.
- **Neutral-corpus PCA denoising**: Some of what warmth and competence have in common in the model

has nothing to do with either concept specifically; it looks more like a general tone the model applies regardless of topic. To find and remove that general tone, we ran 1,500 real Wikipedia articles on ordinary, unrelated topics, text with nothing to do with warmth or competence, through the same model and looked at what the warmth and competence directions have in common with those neutral texts. That shared, non-specific part is then subtracted out of both direction vectors, leaving only what is actually specific to warmth and to competence. The full nine-model procedure, run as a robustness check, is in Tab. S.4.

All 36 pairwise agreement scores are in Tab. S.5; Results shows the same comparison as a heatmap.

**Steering the Concept Vectors.** Finding a direction that separates high- and low-condition stories does not yet prove the model actually uses that direction when forming its judgment. To test that, we push the model's internal activity along the direction while it answers, using activation steering (Turner et al., 2023) via TransformerLens hooks at the probe layer (Fig. 2), and see whether its answer changes.

Because different models' internal activations are different sizes (see the norm column in Tab. 1), we measure how hard we push relative to each model's own typical activation size, not in raw units, so the same push strength means the same thing across all nine models. The push itself is

$$h' = h + \alpha \, \|\bar{h}\| \, \hat{v},$$

where $h$ is the story's original activation, $\hat{v}$ is the unit-normalized target direction, $\|\bar{h}\|$ is that model's mean residual-stream norm from Tab. 1, and $\alpha$ is the strength we choose.

At the concept level, the model answers a direct yes-or-no question: is this protagonist high in warmth (or competence)? Five strengths are tested, $\alpha \in \{-0.10, -0.05, 0, +0.05, +0.10\}$, spanning a strong pull toward "low" to a strong pull toward "high." This range was fixed after an initial sweep that also tested the wider strengths $\alpha \in \{\pm 0.25, \pm 0.50\}$ on Gemma-3-12B and Gemma-3-27B, where the strong settings saturated the response instead of shifting it cleanly: at $\pm 0.50$ the model stops tracking the story and returns Yes or No almost mechanically, the same breakdown at high steering strength that Turner et al. (2023) report for activation steering more generally. Saturation here follows from the structure of the concept prompt, whose answer depends directly on the story the model is asked to read, so a push strong enough to drown out that story yields a fixed answer rather than a graded one. Because that behavior is a property of the task and not of any single checkpoint, we did not repeat the wide sweep on the other seven models and applied the narrow range uniformly to all nine. The full saturation curves are reported in the Results. For each strength, the direction is applied to 10 held-out topics it was never built from, separate from the 40 topics used to build it. For the Gemma-4 and Qwen3.6 checkpoints this yes-or-no question is presented through each model's native chat template, since these more recent chat-tuned checkpoints do not reliably return a clean single-token Yes or No to a raw-text prompt; the other four models receive it as raw text, matching how each checkpoint was run throughout.

We record how much more the model favors Yes over No at each strength, the next-token logit margin:

$$\Delta = \mathrm{logit}(\mathrm{Yes}) - \mathrm{logit}(\mathrm{No}),$$

where logit(Yes) and logit(No) are the model's raw output scores for the two answer tokens before any softmax normalization, so a positive $\Delta$ means the model favors Yes and a larger $\Delta$ means a stronger preference. Throughout, the raw dense directions define the primary causal comparisons reported here; the PCA-denoised vectors serve as a robustness check (Supplementary, PCA Denoising).

A single steering result does not tell us whether the effect is specific to warmth or competence, or whether it would show up for almost any direction we pushed along. To check this, we compare six different directions and see how each one shifts the model's answer:

- **Dense target direction**: The same warmth or competence direction used everywhere else in this study, the difference between the average high- and low-condition activations. This is the main direction we are actually trying to validate.
- GS2 **Decoded Gemma Scope 2 SAE direction**: Gemma Scope 2 (Google DeepMind Language Model Interpretability Team, 2025) is a separate tool from Google DeepMind that breaks a model's internal activity down into thousands of smaller, more specific pieces. We compute the same warmth/competence contrast inside this alternative breakdown, then translate it back into the model's own activation space, to see whether steering still works when built from a completely different representation.
- GS2 **Axis-specific component**: Within that breakdown, some pieces move together for both warmth and competence, and some move only for whichever concept is being tested. This direction keeps only the part specific to the one concept being tested.
- GS2 **Component shared between the two axes**: The opposite of the direction above, only the part that moves for both warmth and competence together, the piece that is not specific to either concept on its own.
- GS2 **Opposing concept axis**: When testing warmth, we steer along the competence direction instead, and vice versa when testing competence, to see

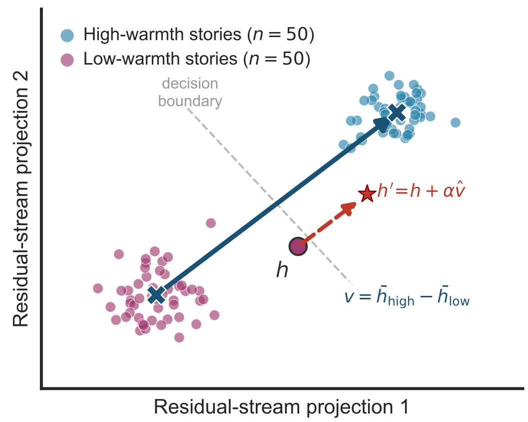


**Figure 2: Residual-stream concept geometry (schematic).** Low-warmth and high-warmth stories occupy different regions of this space, and the mean-difference direction $\boldsymbol{v} = \bar{\boldsymbol{h}}_{\textbf{high}} - \bar{\boldsymbol{h}}_{\textbf{low}}$ (blue arrow) is what separates them. This panel shows only the warmth axis, so its $v$ is the vector written $v_W$ in Methods, with competence $v_C$ defined the same way. Steering acts like grabbing the wheel of a car that is already driving down one street and turning it until the car crosses onto the next street over: the red dashed arrow, $\boldsymbol{h'} = \boldsymbol{h} + \boldsymbol{\alpha}\,\hat{\boldsymbol{v}}$, shows this same kind of forced redirection, pushing a story's representation from the low-warmth region, across the decision boundary, into the high-warmth region. Concretely, this is done by adding a scaled copy of that same high-warmth direction to the story's current representation. Illustrative synthetic activations, not measured data.

whether an unrelated concept's direction moves the answer too.

- **Random orthogonal control**: A direction chosen at random and adjusted so it does not overlap at all with the real target direction, our chance-level baseline.

The dense target direction and a random orthogonal control were both tested on all nine models; the other four were tested only on Gemma-3-12B and Gemma-3-27B. Those four, marked GS2 above, are built from a Gemma Scope 2 decomposition, and that tool covers only the Gemma-3 generation (Google DeepMind Language Model Interpretability Team, 2025), so they could not be extended to Gemma-4, Qwen, or Llama. The nine-model random control is a differently constructed, calibrated set of many random directions per model rather than the single orthogonalized vector compared here, but it serves the same chance-level baseline role. If steering only shifts the answer along the real target direction, and not along the unrelated ones, that supports the effect being specific to warmth or competence rather than an artifact of pushing the residual stream in any direction. Uncertainty throughout is estimated by paired topic bootstrap resampling.

**Steering Hiring Decisions.** The concept-level test shows the model will move its stated judgment when pushed; whether that carries over to an actual hiring decision is a separate question. To answer it, we bring the callback task into the same steering setup. The hiring prompt pairs each of the 282 applicant first names from Gallo et al. (2024) with one fixed administrative-assistant application, identical in education, experience, and skills across every trial, so the name is the only signal that changes. The model reads the application and answers a single question, "would you recommend calling this candidate back for an interview? Answer with a single word: Yes or No." We score that answer with the callback margin,

$$\Delta_{\text{callback}} = \text{logit}(\text{Yes}) - \text{logit}(\text{No}),$$

the same Yes-versus-No logit difference used at the concept level, now read from the hiring prompt rather than the concept-judgment prompt. Demographic labels stay out of the prompt. Race and gender are never shown to the model; each name is tagged afterward from the Gallo et al. (2024) coding, mirroring how human correspondence studies read group membership from names alone (Bertrand and Mullainathan, 2004). We then apply the same steering hooks used at the concept level while the model works through these applications, over a 60-name subset of the roster. Strength is swept broadly at $\alpha \in \{\pm 0.25, \pm 0.50\}$ and refined at $\{\pm 0.05, \pm 0.10\}$ on all nine models, so that a push on the warmth or competence direction can be traced directly to a shift in the callback margin. No high-versus-low split is applied to the names at this stage either; the intervention lives entirely in the signed strength $\alpha$, which varies continuously, while each name only fixes which application the model reads. Unlike the concept judgment, the hiring decision does not saturate under the wider strengths, so both regimes stay usable here. Its callback answer turns on a single applicant name set against a fixed, neutral application, and the model already leans one way before any push, so steering at $\pm 0.50$ scales that existing leaning up or down rather than locking the answer, and the response stays close to linear across the broad grid for roughly a third of model-axis combinations, though several checkpoints depart from it (Results reports the per-model fit). Refining to $\pm 0.05$ and $\pm 0.10$ then resolves the smaller, near-baseline shifts that the wider grid steps over. Per-model steering curves and slopes are reported in the Results.

**Names, Human Ratings, and Disparity.** Beyond the causal steering test, we ask whether the underlying representations track real human perception and real-world demographic gaps, using four analyses:

- **Probe-versus-human alignment**: Independently of the hiring prompt, each applicant name is embedded in the neutral carrier sentence "The job applicant's name is {Name}.", and the resulting residual-stream activation is projected onto the raw warmth and competence directions to yield a per-name probe score; no denoised counterpart is computed at this stage. We compare these scores against the aggregated human ratings behind them (16,557 rater judgments from 787 raters across nine source studies) using Spearman rank correlation, which tells us whether the model's internal sense of a name's warmth or competence lines up with how people actually perceive that name. Each name already carries a continuous crowdsourced warmth and competence score, so no high-versus-low grouping is imposed here, unlike the concept stories used to build the directions. The correlation runs directly on the graded ratings, since sorting names into poles would discard information the continuous scale already carries.

- **Group-level callback disparity**: Callback disparities are computed as differences in mean callback margin between name groups (race: 47 Black-signaling versus 180 White-signaling names; gender: 154 female versus 115 male), standardized by the within-model standard deviation of the callback margin so that models with different logit scales can be compared on equal footing. For the Gemma-4 and Qwen3.6 checkpoints the hiring prompt is presented through each model's native chat template, since these more recent chat-tuned checkpoints do not reliably return a clean single-token Yes or No to a raw-text prompt; the other five models receive it as raw text.
- **Human reference gaps**: These come from the published per-name callback rates, matched by first name and source study rather than first name alone, since the same first name can carry a different callback rate in each source study. A name rated under more than one study is matched separately against each study's own rate rather than blended into a single value. Some source studies (Neumark, Farber) also vary applicant age as an experimental factor; since our own name ratings carry no age dimension, callback rates are averaged across age within each name-study pair before matching. In total, 246 name-study observations across 186 distinct names match the benchmark this way.
- **Bootstrap mediation**: Finally, we test whether the internal warmth and competence representations actually carry the name-group signal into the callback decision, using bootstrap mediation along the path name group $\rightarrow$ probe score $\rightarrow$ callback margin (5,000 resamples, seed 20260527, $n = 269$ names with complete data, race subset $n = 227$), reporting indirect effects with 95% confidence intervals. Results reports this procedure for all nine models and both race and gender groupings, thirty-six combinations in total.

# Results

**Vector Validation, All Nine Models.** Every one of the five single-model checks and two cross-model analyses described in Methods was run on all nine checkpoints, and none flags a direction as an artifact of the specific stories used to build it.

Tab. 2 collects four of the five single-model checks; 5-fold cross-validated and topic-holdout accuracy are not columns in that table because both reach exactly 1.00 for every model on both axes and every fold, so there is no variation left to tabulate. Cohen's $d$ ranges from 2.70 (Gemma-3-12B, warmth) to 9.97 (Qwen3-14B, competence), and the random-direction null comparison never comes close: across all nine models and both axes, 0 of the 1,000 random directions tested per model matched or exceeded the real direction's separation, with $z$-scores between 3.7 and 15.1. Split-half cosine stability stays in the 0.69-0.90 range reported in Methods, and calibrated cross-axis accuracy stays between 0.82 and 1.00, confirming that warmth and competence share substantial evaluative content rather than behaving as fully separate constructs.

Agreement across architectures is just as consistent. Spearman correlation on the 36 possible model pairs never turns negative: overall ranking agreement (which includes the high-versus-low split itself) ranges from 0.741 to 0.992 across both axes, and the harder within-condition comparison, which asks whether two models rank stories the same way inside one condition rather than just separating high from low, ranges from 0.095 to 0.899 with a median near 0.46 for warmth and 0.54 for competence. Fig. 4 shows the full 9x9 pattern; the complete 36-pair table is in Tab. S.5.

Neutral-corpus PCA denoising, reported in full in Tab. S.4, shows the same pattern across every model: removing the components shared with 1,500 ordinary Wikipedia paragraphs lowers the warmth-competence cosine but never drives it to zero. The reduction is largest in the two Gemma-3 checkpoints (0.749 to 0.530 and 0.708 to 0.487) and comparatively small elsewhere, typically a few hundredths, suggesting that most of what remains after denoising is genuine conceptual overlap rather than a removable general tone.

| Model | Cohen's $d$ (W / C) | Random-null $z$ (W / C) | Split-half cosine (W / C) | Cross-axis accuracy (W→C / C→W) |
|---|---|---|---|---|
| Gemma-3-12B | 2.70 / 2.83 | 3.9 / 3.7 | 0.82 / 0.90 | 0.87 / 0.82 |
| Gemma-3-27B | 2.95 / 3.27 | 4.3 / 4.5 | 0.81 / 0.88 | 0.90 / 0.86 |
| Llama-3.1-8B | 8.48 / 9.07 | 15.0 / 15.1 | 0.70 / 0.83 | 0.99 / 1.00 |
| Gemma-4-12B | 8.63 / 9.04 | 14.1 / 12.8 | 0.77 / 0.87 | 1.00 / 0.98 |
| Gemma-4-26B-A4B | 8.36 / 8.75 | 15.0 / 14.3 | 0.69 / 0.85 | 1.00 / 1.00 |
| Gemma-4-31B | 7.56 / 6.03 | 13.1 / 8.6 | 0.73 / 0.86 | 1.00 / 0.95 |
| Qwen3-14B | 8.97 / 9.97 | 14.1 / 14.6 | 0.71 / 0.85 | 1.00 / 1.00 |
| Qwen3.6-27B | 7.98 / 8.99 | 12.9 / 12.9 | 0.76 / 0.86 | 0.99 / 1.00 |
| Qwen3.6-35B-A3B | 6.31 / 7.35 | 11.3 / 10.4 | 0.71 / 0.86 | 1.00 / 0.97 |

**Table 2: Probe validation, all nine models.** Warmth (W) and competence (C) values in each cell. Cohen's $d$ is the effect size separating high- and low-condition stories on the raw dense direction. Random-null $z$ standardizes that same $d$ against 1,000 random directions' own high/low separation on the same stories; in every model and axis, 0 of 1,000 random directions matched or exceeded the real direction. Split-half cosine is the cosine between two directions independently built from non-overlapping 25-story halves of each condition. Cross-axis accuracy (calibrated) is how well a story's projection onto one axis's direction predicts its label on the other axis. 5-fold cross-validated and topic-holdout accuracy are omitted here since both are 1.00 for every model on both axes.

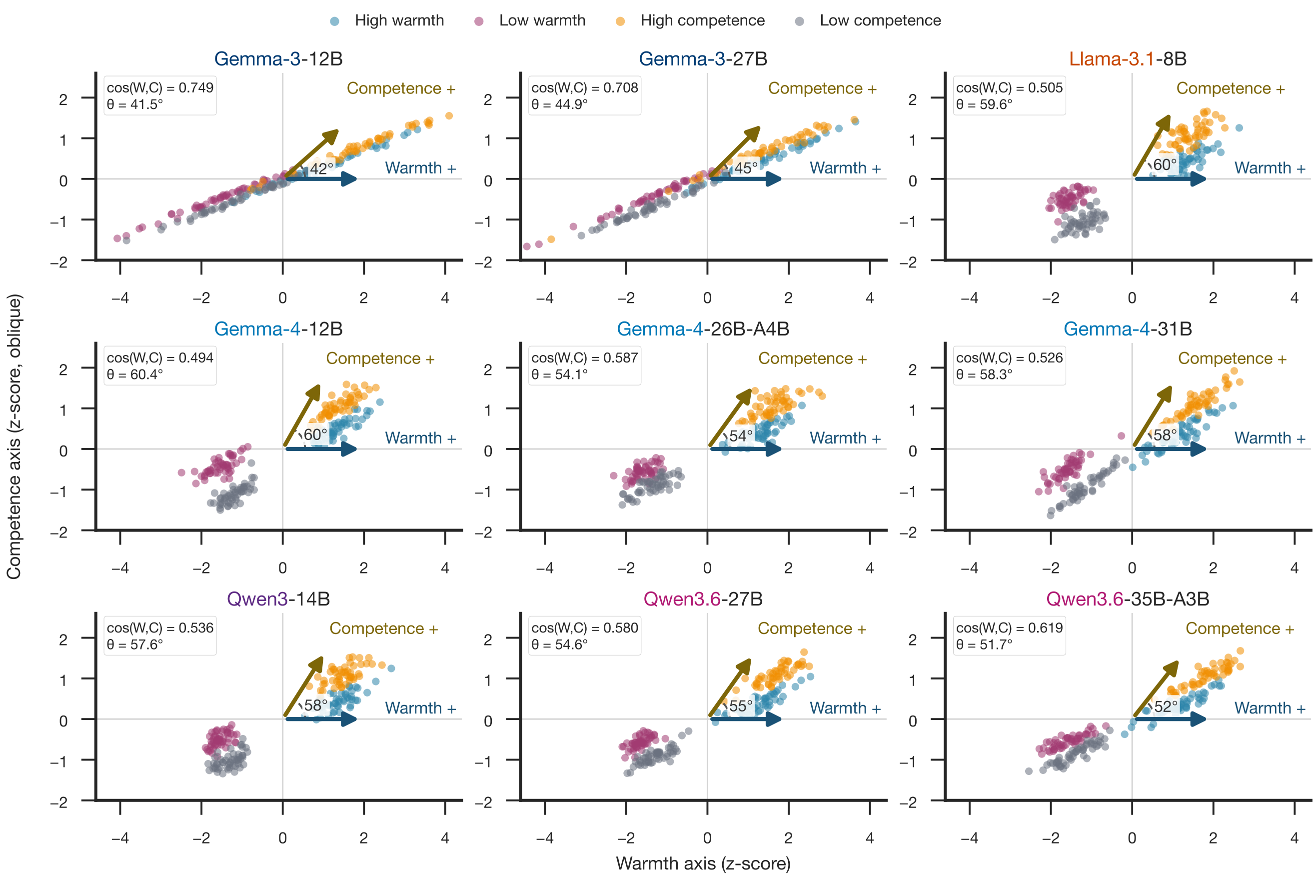


**Figure 3: Warmth and competence directions share model-dependent geometry.** Each panel projects the same 200 synthetic stories onto raw dense probe-layer directions in an oblique basis. The arrow angle equals $\mathbf{arccos[cos(}\boldsymbol{v_W, v_C}\mathbf{)]}$, so smaller angles indicate stronger geometric alignment. All panels use model-internal z-scores, shared axis limits, and equal x/y scaling.

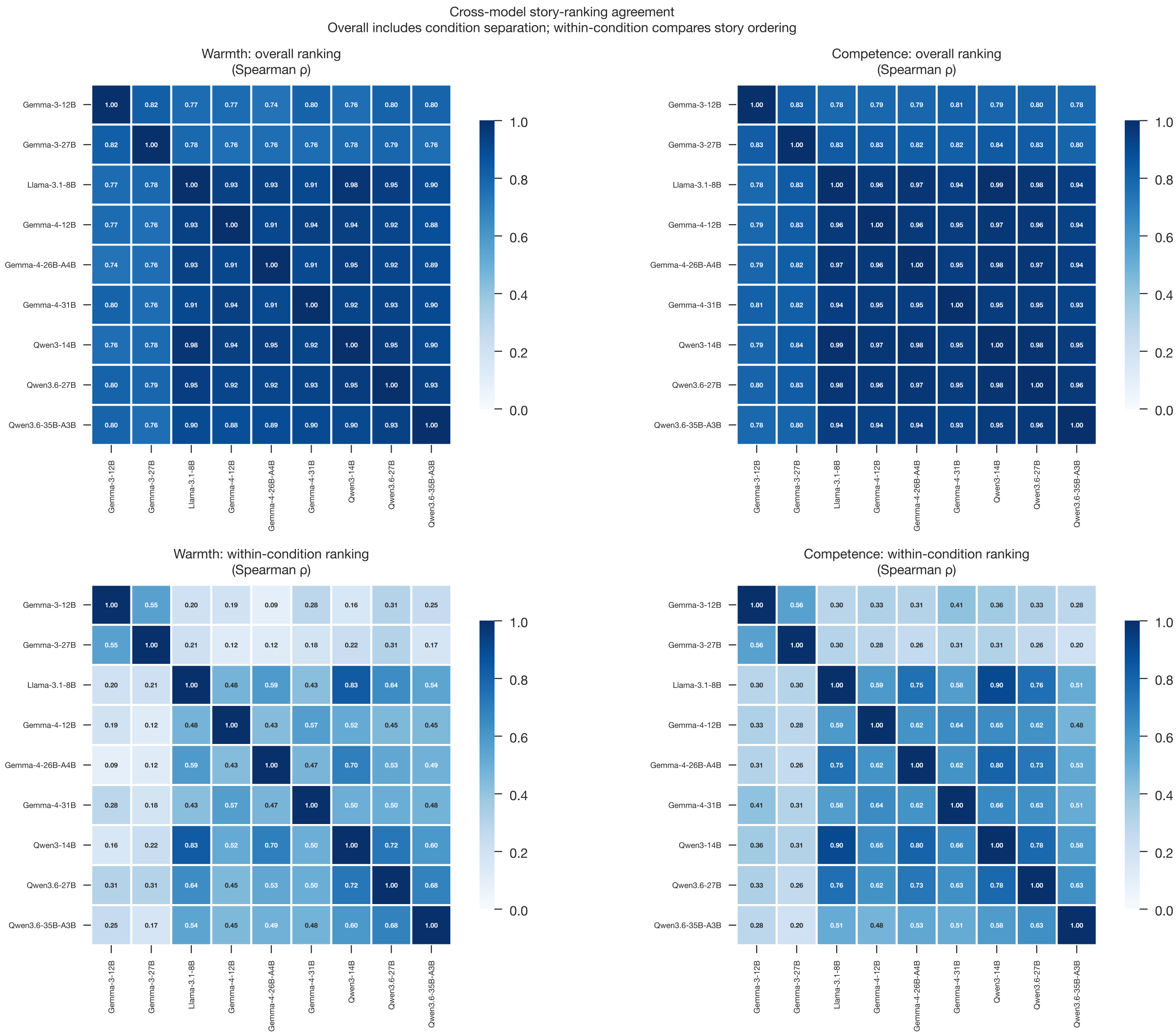


**Figure 4: Cross-model story-ranking agreement, all nine models.** Each cell is a Spearman correlation between two models' projections of the same 200 stories onto their own warmth or competence direction. Overall (top row) includes the high-versus-low condition split; within-condition (bottom row) restricts the comparison to stories sharing one condition, so it isolates agreement on finer story ordering. Full pairwise values are in Tab. S.5.

**Steering the Concept Vectors, All Nine Models.** At the narrow local grid ($\alpha \in \{-0.10, ..., +0.10\}$), Fig. 5 tracks the held-out concept margin against steering strength in every model, normalized by each model's own baseline high-low gap so the curves stay comparable despite very different raw logit scales. Gemma-3-12B reaches the largest positive endpoint on both axes (0.236 warmth, 0.141 competence), Qwen3-14B is second on both (0.122, 0.104), and Gemma-4-26B-A4B and Gemma-4-31B stay nearly flat at the competence endpoint (0.001 and 0.008) despite separating high-competence from low-competence stories cleanly in the checks above. A model can encode a concept strongly while still responding weakly to an additive push along its own mean-difference direction.

Wide strengths tell a partly different story. Tab. 3 sweeps $\alpha \in \{\pm 0.25, \pm 0.50\}$ on Gemma-3-12B and Gemma-3-27B, the two models tested at this range. Three of the four model-axis rows show the step from +0.25 to +0.50 shrinking relative to the step before it, matching the saturating breakdown Methods describes for extreme strengths; Gemma-3-27B's warmth response is the exception, still accelerating at +0.50 rather than leveling off.

A single direction moving the answer does not by itself establish that the effect is specific to warmth or competence rather than any sufficiently large push. Tab. 4 compares six directions at the local endpoint on the same two models. The dense target direction produces the row's largest effect in only one of the four rows (Gemma-3-12B warmth); in the other three, an unrelated direction, most often the SAE-decoded or axis-specific component, matches or exceeds it. This does not read as clean specificity, and points instead toward the same shared evaluative content the neutral-corpus PCA denoising already found between warmth and competence. Two further checks, reported in the Supplementary Materials, extend this picture: feature ablation (Tab. S.7) disrupts the high-low margin gap more when the features shared between axes are removed than when the axis-specific features are, in three of four rows, and cross-scale feature matching (Tab. S.8) finds Gemma-3-12B's and Gemma-3-27B's SAE features correlate well above a permutation null (Tab. S.6 reports the underlying reconstruction quality behind both checks).

Comparing the dense target against a random-direction control extends across all nine models, but the control itself is not built the same way everywhere. Tab. 5 reports both effects for every model and axis, with the control's basis stated in its own column: five checkpoints (Gemma-4-12B, Gemma-4-26B-A4B, Gemma-4-31B, Qwen3.6-27B, Qwen3.6-35B-A3B) carry a calibrated set of 99 SD-matched random directions, while the four original checkpoints carry a single random direction. Where the calibrated control exists, the result splits by family: both Qwen3.6 checkpoints place the dense-target effect clearly outside the random control's interval, while all three Gemma-4 checkpoints have a random interval wide enough to contain the dense-target effect, so this comparison alone does not establish specificity for that family. Extending the 99-direction control to the four legacy checkpoints needs new GPU runs and is deliberately left for later rather than closed here.

**Steering Hiring Decisions, All Nine Models.** Pushing the warmth or competence direction while the model reads a hiring application shifts the callback margin $\Delta_{\text{callback}}$ in every model, tested on the same 60-name subset at the broad strength grid $\alpha \in \{\pm 0.25, \pm 0.50\}$, chosen here uniformly across all nine models so the comparison sits on one shared scale rather than mixing the per-model preferred endpoints used by Tab. S.9.

Tab. 6 fits a line through each model-axis response and reports the slope alongside $R^2$. The fit is close to linear in only 7 of the 18 rows, Gemma-3-12B warmth, both Llama-3.1-8B axes, Gemma-4-26B-A4B warmth, and both Qwen3.6-27B and Qwen3.6-35B-A3B warmth axes; the remaining eleven depart from a straight line, most visibly Gemma-3-27B, where $R^2$ stays below 0.35 on both axes and the endpoint delta at $\alpha = +0.50$ is actually negative despite a positive slope estimate. Slopes vary in sign and size as well as fit quality: Gemma-3-12B warmth has the steepest response (+12.91 per unit $\alpha$), and Qwen3-14B competence is the only row with a negative slope (-0.11), effectively flat. The categorical Yes/No transitions behind this same steering intervention, including how often it crosses the decision boundary rather than only scaling an existing lean, are reported in full in Tab. S.9.

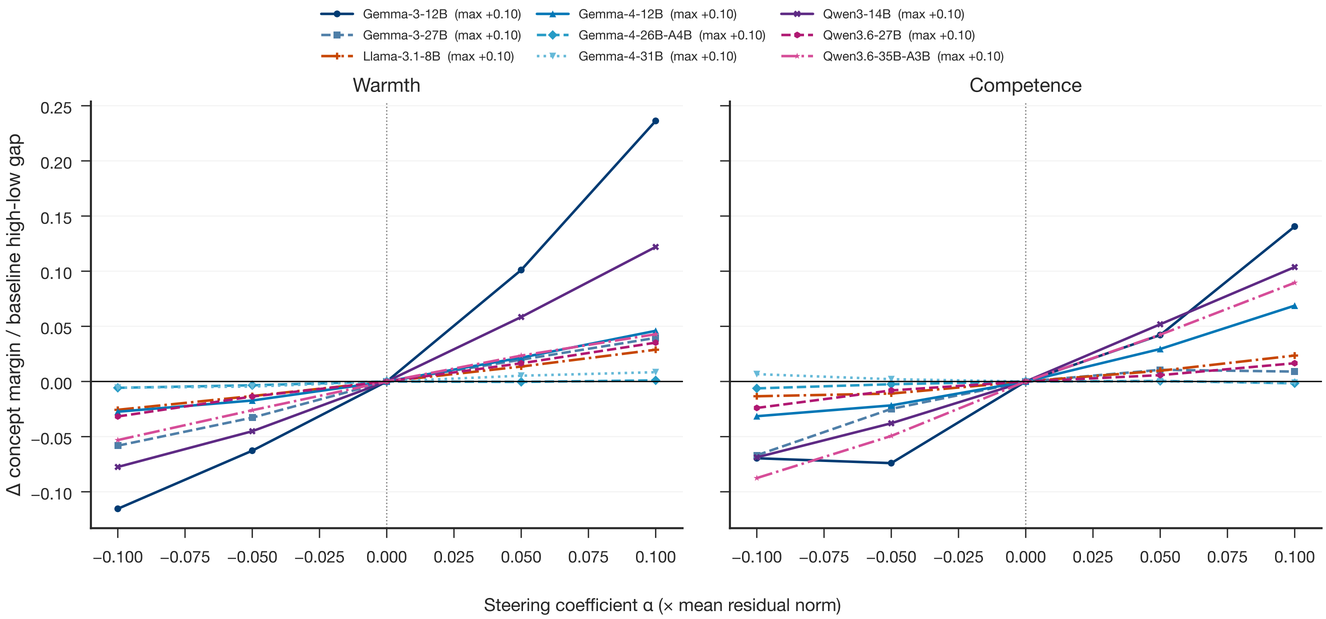


**Figure 5: Normalized concept-level steerability varies across nine checkpoints.** Each curve shows the change in the held-out concept margin $\mathbf{\Delta}$ (the Yes-versus-No logit difference defined in Methods) produced by target-direction steering, divided by the baseline high-versus-low gap in $\mathbf{\Delta}$ for the same model and axis. Parentheses in the legend report the maximum positive steering coefficient, $\alpha = +0.10$, expressed relative to the mean residual norm. Normalization reduces raw logit-scale differences but does not establish direction specificity; calibrated random and cross-axis controls are interpreted in the text.

| Model | Axis | $\alpha = -0.50$ | $-0.25$ | 0 | $+0.25$ | $+0.50$ |
|---|---|---|---|---|---|---|
| Gemma-3-12B | Warmth | -5.68 | -3.30 | +0.00 | +7.29 | +11.43 |
| Gemma-3-12B | Competence | -4.09 | +0.79 | +0.00 | +8.70 | +7.32 |
| Gemma-3-27B | Warmth | +0.03 | -0.96 | +0.00 | +6.17 | +14.44 |
| Gemma-3-27B | Competence | +4.28 | +1.23 | +0.00 | +3.68 | +5.08 |

**Table 3: Concept-level steering at wide strengths.** Change in the held-out Yes-versus-No logit margin under dense-target steering, Gemma-3-12B and Gemma-3-27B, the two models where the wide strength grid was tested. The step from +0.25 to +0.50 is smaller than the step before it for three of the four rows, consistent with the saturating breakdown Methods describes; Gemma-3-27B warmth is the exception, still accelerating at +0.50.

| Model | Axis | Dense target | SAE decoded | Axis-specific | Shared | Opposing axis | Random |
|---|---|---|---|---|---|---|---|
| Gemma-3-12B | Warmth | **+3.88** | +2.30 | +2.10 | +2.16 | +3.24 | -0.32 |
| Gemma-3-12B | Competence | +2.01 | +3.69 | +3.12 | +2.34 | +2.38 | +2.88 |
| Gemma-3-27B | Warmth | +1.02 | -0.19 | -1.38 | +0.69 | +2.86 | +1.34 |
| Gemma-3-27B | Competence | +0.21 | +3.53 | +4.36 | +1.61 | +0.31 | +2.67 |

**Table 4: Direction specificity at the local endpoint ($\alpha = +0.10$).** Change in the held-out Yes-versus-No logit margin for six directions, Gemma-3-12B and Gemma-3-27B. Dense target is the same mean-difference direction used everywhere else in this study; SAE decoded, axis-specific, and shared are built from a Gemma Scope 2 sparse decomposition; opposing axis steers along the other concept's direction; random is a single orthogonalized control. Bold marks the row's largest effect. Dense target is largest in only 1 of the four model-axis rows; the other three are matched or exceeded by at least one unrelated direction, most often the SAE-decoded or axis-specific component, which does not support a clean specificity reading and is discussed in the text alongside the shared warmth-competence overlap already noted in PCA denoising.

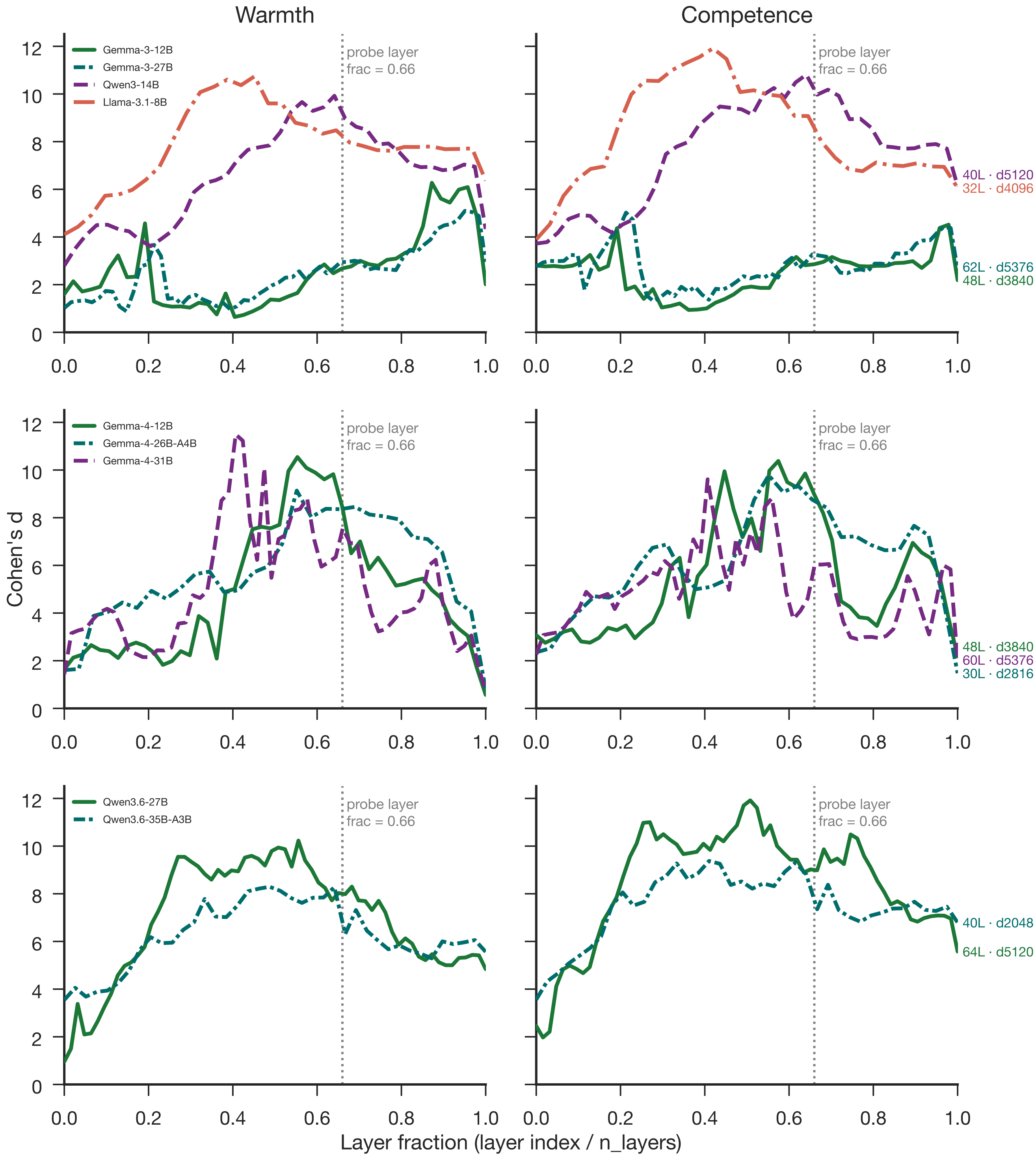


**Figure 6: Warmth and competence directions emerge robustly across depth.** Layer-wise Cohen's $d$ across nine open-weights models. The dotted vertical line marks the fixed probe-layer fraction of 0.66.

| Model | Axis | Dense target ($\alpha = +0.10$) | Random control | Control basis |
|---|---|---|---|---|
| Gemma-3-12B | Warmth | +3.81 | +0.10 [-0.32, +0.49] | 1 direction |
| Gemma-3-12B | Competence | +1.98 | +0.30 [+0.05, +0.56] | 1 direction |
| Gemma-3-27B | Warmth | +1.02 | +0.61 [+0.32, +0.89] | 1 direction |
| Gemma-3-27B | Competence | +0.21 | -3.36 [-4.00, -2.76] | 1 direction |
| Llama-3.1-8B | Warmth | +0.26 | +0.06 [+0.03, +0.10] | 1 direction |
| Llama-3.1-8B | Competence | +0.18 | -0.12 [-0.14, -0.11] | 1 direction |
| Gemma-4-12B | Warmth | +1.84 | +0.29 ± 2.77 | 99 SD-matched dirs. |
| Gemma-4-12B | Competence | +2.66 | +0.23 ± 2.56 | 99 SD-matched dirs. |
| Gemma-4-26B-A4B | Warmth | +0.05 | -0.84 ± 2.94 | 99 SD-matched dirs. |
| Gemma-4-26B-A4B | Competence | -0.07 | -0.66 ± 2.71 | 99 SD-matched dirs. |
| Gemma-4-31B | Warmth | +0.44 | +0.85 ± 3.16 | 99 SD-matched dirs. |
| Gemma-4-31B | Competence | -0.06 | +0.78 ± 2.54 | 99 SD-matched dirs. |
| Qwen3-14B | Warmth | +1.20 | -0.02 [-0.10, +0.06] | 1 direction |
| Qwen3-14B | Competence | +1.11 | +0.23 [+0.17, +0.28] | 1 direction |
| Qwen3.6-27B | Warmth | +0.69 | +0.01 ± 0.04 | 99 SD-matched dirs. |
| Qwen3.6-27B | Competence | +0.31 | +0.01 ± 0.03 | 99 SD-matched dirs. |
| Qwen3.6-35B-A3B | Warmth | +0.82 | +0.00 ± 0.06 | 99 SD-matched dirs. |
| Qwen3.6-35B-A3B | Competence | +1.45 | -0.02 ± 0.06 | 99 SD-matched dirs. |

**Table 5: Dense-target steering versus a random-direction control, all nine models.** The random-control basis is not uniform: Gemma-4-12B, Gemma-4-26B-A4B, Gemma-4-31B, Qwen3.6-27B, and Qwen3.6-35B-A3B use 99 SD-matched random directions per model, reported as mean ± 95% CI; Gemma-3-12B, Gemma-3-27B, Llama-3.1-8B, and Qwen3-14B use a single random direction, reported with its own bootstrap CI. Extending the calibrated 99-direction control to the remaining four models requires new GPU runs and is left for future work rather than closed here.

| Model | Axis | Slope ($\Delta$ per unit $\alpha$) | $R^2$ | $\Delta$ at $\alpha = +0.50$ |
|---|---|---|---|---|
| Gemma-3-12B | Warmth | +12.91 | **0.92** | +8.35 |
| Gemma-3-12B | Competence | +9.07 | 0.66 | +6.23 |
| Gemma-3-27B | Warmth | +1.16 | 0.03 | -0.23 |
| Gemma-3-27B | Competence | +2.96 | 0.35 | -0.98 |
| Llama-3.1-8B | Warmth | +3.97 | **0.89** | +3.17 |
| Llama-3.1-8B | Competence | +2.50 | **0.86** | +2.17 |
| Gemma-4-12B | Warmth | +8.76 | 0.65 | -1.26 |
| Gemma-4-12B | Competence | +6.48 | 0.65 | -0.13 |
| Gemma-4-26B-A4B | Warmth | +3.60 | **0.91** | +0.75 |
| Gemma-4-26B-A4B | Competence | +0.65 | 0.06 | -1.99 |
| Gemma-4-31B | Warmth | +7.83 | 0.51 | -2.59 |
| Gemma-4-31B | Competence | +3.07 | 0.21 | -3.64 |
| Qwen3-14B | Warmth | +1.85 | 0.80 | +0.60 |
| Qwen3-14B | Competence | -0.11 | 0.07 | -0.38 |
| Qwen3.6-27B | Warmth | +5.69 | **0.94** | +2.24 |
| Qwen3.6-27B | Competence | +4.87 | **0.92** | +1.07 |
| Qwen3.6-35B-A3B | Warmth | +3.73 | **0.91** | +1.23 |
| Qwen3.6-35B-A3B | Competence | +1.45 | 0.30 | -1.09 |

**Table 6: Hiring callback steering, broad grid, all nine models.** Mean change in the callback margin $\Delta_{\text{callback}}$ across the 60-name subset, at $\alpha \in \{\pm 0.25, \pm 0.50\}$. Slope and $R^2$ come from an ordinary least-squares fit of the five per-strength means; bold marks $R^2 \geq 0.8$. The fit is close to linear in only 7 of the 18 model-axis rows; several others, most visibly Gemma-3-27B on both axes, show weak or non-monotonic trends instead.

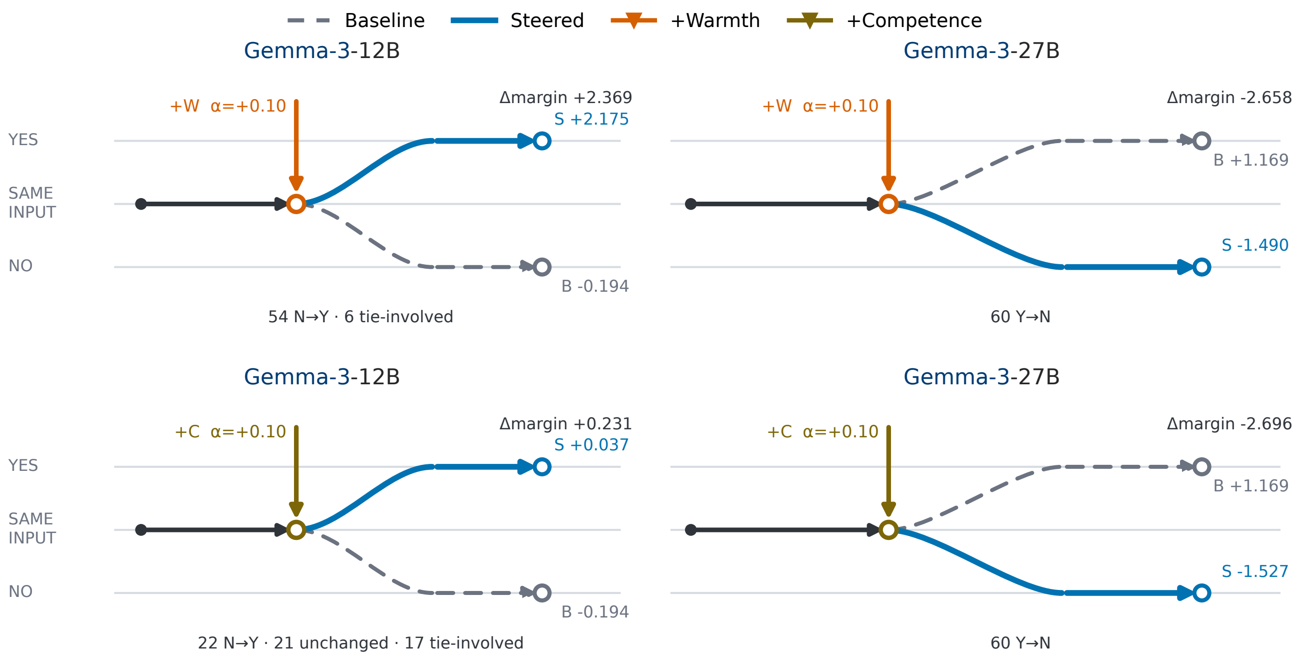


**Figure 7: Matched examples show that positive concept steering can move callback decisions in either direction.** Rows show warmth and competence; columns compare Gemma-3-12B, whose mean decision moves from No to Yes, with Gemma-3-27B, whose mean decision moves from Yes to No. All panels use the same 60 applications and $\alpha = +0.10$. Gray dashed paths show baseline outcomes, blue paths show steered outcomes, and bottom annotations summarize individual-name transitions. The panels were selected to display both transition directions under matched conditions, not to estimate their prevalence. Endpoint margins, strengths, and transition counts are empirical; connector geometry is schematic. Complete results for all nine models and both axes appear in Tab. S.9.

**Names, Human Ratings, and Disparity, All Nine Models.** Beyond the causal steering test, four analyses ask whether the probed representations track real human perception and real-world demographic callback gaps.

Independently of the hiring prompt, each of the 282 rated applicant names is embedded in the neutral carrier sentence "The job applicant's name is {Name}.", and the resulting residual-stream activation is projected onto the raw warmth and competence directions to yield a per-name probe score. Tab. 7 correlates these scores with the aggregated human ratings behind them (16,557 rater judgments from 787 raters across nine source studies) using Spearman rank correlation. Warmth alignment ranges from $\rho = -0.287$ (Llama-3.1-8B) to +0.388 (Gemma-3-27B); competence alignment ranges from $\rho = -0.058$, not significant (Llama-3.1-8B), to +0.466 (Qwen3-14B). Competence tracks human perception more consistently than warmth: four of the nine models show a warmth correlation that is either negative or not significant, while every model's competence correlation is positive, and all but one reach significance. The same table's callback columns report a separate quantity, the correlation between each model's own probe projection and its own unsteered callback margin, which is not a human-alignment measure.

Tab. 8 asks the same alignment question at the group level: does a group's model-internal warmth or competence lean the same direction as that group's human rating? Race (Black/White) and gender (Female/Male) are treated as separate marginal axes, following Gallo et al. (2024)'s own grouping convention. Model warmth/competence and human warmth/competence live on different raw scales (human ratings are 0-100 Likert averages, model values are unbounded residual-stream projections), so both are standardized to $z$-scores against the full 282-name distribution before being placed side by side; a positive value in both the model and human column for a group means the two lean the same direction. Raw, study-broken values are in Tab. S.10. The human-side numbers in this table come from published per-name callback rates, matched by first name and source study rather than by first name alone, since the same first name can carry a different callback rate in each source study; averaging across age where a study varies it yields 246 name-study observations across 186 distinct names.

Tab. 9 crosses race and gender into four joint groups (Black-Female, Black-Male, White-Female, White-Male), recovering the interaction structure the marginal breakdown above collapses, using the same standardized comparison. The study-broken raw values behind this table are in Tab. S.11.

The last analysis asks whether the probed representations actually carry the name-group signal into the callback decision, rather than merely correlating with it. Tab. 10 tests this with bootstrap mediation along the path name group → probe score → callback margin, run separately for race and gender and for warmth and competence, thirty-six tests across the nine models. Fourteen of the thirty-six indirect effects exclude zero at the uncorrected 95% level; under a Bonferroni threshold across all thirty-six ($\alpha = 0.05/36$), only the Llama-3.1-8B race–warmth path survives (IE = +0.190, 95% CI $[+0.111, +0.292]$), so every other entry should be read as suggestive rather than confirmed. The two Gemma-3 checkpoints show no significant mediation path on either grouping or probe, consistent with representations that steer cleanly but do not themselves drive the hiring decision. The later checkpoints complicate that picture: all three Gemma-4 models show a significant race–competence indirect effect, Gemma-4-26B-A4B additionally shows a significant gender–warmth path, and the two Qwen3.6 checkpoints show a comparable spread of significant warmth and competence paths, matching the pattern already present in Qwen3-14B. Read together, the absence of mediation looks like a property of the Gemma-3 generation specifically rather than a general feature of models whose concept directions steer well.

| Model | Probe layer (frac) | $\rho$(warmth, human) | $\rho$(competence, human) | $\rho$(callback, warmth) | $\rho$(callback, competence) |
|---|---|---|---|---|---|
| Gemma-3-12B | 31 (0.65) | +0.357*** | +0.237*** | +0.150* | +0.149* |
| Gemma-3-27B | 40 (0.65) | +0.388*** | +0.271*** | -0.160** | -0.156** |
| Llama-3.1-8B | 20 (0.62) | -0.287*** | -0.058 n.s. | +0.092 n.s. | +0.053 n.s. |
| Gemma-4-12B | 31 (0.65) | +0.009 n.s. | +0.216*** | -0.110 n.s. | -0.124* |
| Gemma-4-26B-A4B | 19 (0.63) | -0.190** | -0.044 n.s. | +0.510*** | +0.373*** |
| Gemma-4-31B | 39 (0.65) | +0.254*** | +0.290*** | +0.024 n.s. | +0.337*** |
| Qwen3-14B | 26 (0.65) | -0.178** | +0.466*** | +0.195** | +0.426*** |
| Qwen3.6-27B | 42 (0.66) | +0.192** | +0.247*** | +0.302*** | +0.220*** |
| Qwen3.6-35B-A3B | 26 (0.65) | +0.206*** | +0.130* | +0.344*** | +0.197*** |

**Table 7: Name-level correlation between probed warmth/competence and human ratings.** For each of the 282 rated applicant names, the model's residual-stream activation at the probe layer is projected onto the warmth and competence direction vectors. "Probe layer" gives the selected layer index and its resolved fraction of total depth (targeted probe_layer_frac = 0.66; the reported fraction is the integer-rounded layer divided by model depth). Warmth/competence columns correlate the probe projection with human warmth/competence ratings; callback columns correlate the probe projection with the model's own unsteered callback margin. All correlations are Spearman $\rho$. $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$, n.s. not significant.

| Model | Group | $n$ | Model warmth/competence ($z$) | Human warmth/competence ($z$) | Human callback | Model margin |
|---|---|---|---|---|---|---|
| Gemma-3-12B | Black | 47 | -0.44 / -0.43 | -0.30 / -0.58 | 0.183 | -0.213 |
| | White | 180 | +0.45 / +0.44 | +0.21 / +0.15 | 0.171 | -0.199 |
| | Female | 154 | +0.11 / +0.10 | +0.17 / -0.00 | 0.145 | -0.165 |
| | Male | 115 | -0.04 / -0.03 | -0.16 / +0.07 | 0.181 | -0.258 |
| Gemma-3-27B | Black | 47 | -0.39 / -0.36 | -0.30 / -0.58 | 0.183 | 1.665 |
| | White | 180 | +0.35 / +0.36 | +0.21 / +0.15 | 0.171 | 1.121 |
| | Female | 154 | +0.11 / +0.11 | +0.17 / -0.00 | 0.145 | 1.118 |
| | Male | 115 | -0.08 / -0.08 | -0.16 / +0.07 | 0.181 | 1.312 |
| Llama-3.1-8B | Black | 47 | +0.45 / +0.24 | -0.30 / -0.58 | 0.183 | -2.020 |
| | White | 180 | -0.38 / -0.22 | +0.21 / +0.15 | 0.171 | -2.067 |
| | Female | 154 | +0.00 / +0.18 | +0.17 / -0.00 | 0.145 | -2.054 |
| | Male | 115 | -0.05 / -0.23 | -0.16 / +0.07 | 0.181 | -2.107 |
| Gemma-4-12B | Black | 47 | -0.34 / -0.40 | -0.30 / -0.58 | 0.183 | 17.194 |
| | White | 180 | +0.24 / +0.44 | +0.21 / +0.15 | 0.171 | 17.172 |
| | Female | 154 | -0.26 / +0.01 | +0.17 / -0.00 | 0.145 | 17.208 |
| | Male | 115 | +0.39 / +0.07 | -0.16 / +0.07 | 0.181 | 17.175 |
| Gemma-4-26B-A4B | Black | 47 | +0.56 / +0.36 | -0.30 / -0.58 | 0.183 | 21.509 |
| | White | 180 | -0.37 / -0.39 | +0.21 / +0.15 | 0.171 | 21.095 |
| | Female | 154 | +0.15 / -0.13 | +0.17 / -0.00 | 0.145 | 21.464 |
| | Male | 115 | -0.27 / +0.02 | -0.16 / +0.07 | 0.181 | 20.902 |
| Gemma-4-31B | Black | 47 | -0.12 / -0.19 | -0.30 / -0.58 | 0.183 | 25.808 |
| | White | 180 | +0.30 / +0.36 | +0.21 / +0.15 | 0.171 | 25.577 |
| | Female | 154 | -0.06 / +0.04 | +0.17 / -0.00 | 0.145 | 25.788 |
| | Male | 115 | +0.13 / +0.02 | -0.16 / +0.07 | 0.181 | 25.408 |
| Qwen3-14B | Black | 47 | +0.35 / -0.34 | -0.30 / -0.58 | 0.183 | 1.758 |
| | White | 180 | -0.46 / +0.17 | +0.21 / +0.15 | 0.171 | 1.700 |
| | Female | 154 | +0.13 / +0.19 | +0.17 / -0.00 | 0.145 | 1.851 |
| | Male | 115 | -0.28 / -0.24 | -0.16 / +0.07 | 0.181 | 1.537 |
| Qwen3.6-27B | Black | 47 | +0.13 / -0.24 | -0.30 / -0.58 | 0.183 | 5.402 |
| | White | 180 | -0.02 / +0.15 | +0.21 / +0.15 | 0.171 | 5.328 |
| | Female | 154 | -0.04 / -0.29 | +0.17 / -0.00 | 0.145 | 5.384 |
| | Male | 115 | +0.00 / +0.34 | -0.16 / +0.07 | 0.181 | 5.284 |
| Qwen3.6-35B-A3B | Black | 47 | -0.15 / +0.09 | -0.30 / -0.58 | 0.183 | 2.981 |
| | White | 180 | +0.31 / +0.27 | +0.21 / +0.15 | 0.171 | 3.039 |
| | Female | 154 | +0.24 / +0.14 | +0.17 / -0.00 | 0.145 | 3.143 |
| | Male | 115 | -0.24 / -0.09 | -0.16 / +0.07 | 0.181 | 2.873 |

**Table 8: Warmth/competence bias by marginal demographic group, model versus human, standardized.** Race (Black/White) and gender (Female/Male) are treated as separate marginal axes, following Gallo and Hausladen's own grouping convention. "Model margin" is the model's own Yes/No logit difference on the hiring prompt (unsteered); "Human callback" is the real observed callback rate from the correspondence-study benchmark. Model and human warmth/competence are each standardized to $z$-scores against the full 282-name distribution (SD above or below the overall mean), the same convention already used for callback-margin standardization elsewhere in this paper, so a positive value in both the model and human column for a group indicates the model's internal representation leans the same direction as human perception; raw (unstandardized) values are given in Tab. S.10.

| Model | Race × Gender | $n$ | Model warmth/competence ($z$) | Human warmth/competence ($z$) | Human callback | Model margin |
|---|---|---|---|---|---|---|
| Gemma-3-12B | Black-Female | 28 | -0.46 / -0.47 | -0.24 / -0.66 | 0.175 | -0.183 |
| | Black-Male | 28 | -0.14 / -0.14 | -0.37 / -0.41 | 0.178 | -0.290 |
| | White-Female | 117 | +0.65 / +0.64 | +0.23 / +0.02 | 0.148 | -0.169 |
| | White-Male | 73 | +0.42 / +0.41 | +0.03 / +0.03 | 0.216 | -0.276 |
| Gemma-3-27B | Black-Female | 28 | -0.56 / -0.52 | -0.24 / -0.66 | 0.175 | 1.621 |
| | Black-Male | 28 | -0.19 / -0.15 | -0.37 / -0.41 | 0.178 | 1.781 |
| | White-Female | 117 | +0.50 / +0.51 | +0.23 / +0.02 | 0.148 | 1.019 |
| | White-Male | 73 | +0.22 / +0.23 | +0.03 / +0.03 | 0.216 | 1.293 |
| Llama-3.1-8B | Black-Female | 28 | +0.62 / +0.70 | -0.24 / -0.66 | 0.175 | -1.964 |
| | Black-Male | 28 | +0.47 / -0.17 | -0.37 / -0.41 | 0.178 | -2.049 |
| | White-Female | 117 | -0.31 / -0.13 | +0.23 / +0.02 | 0.148 | -2.060 |
| | White-Male | 73 | -0.39 / -0.43 | +0.03 / +0.03 | 0.216 | -2.096 |
| Gemma-4-12B | Black-Female | 28 | -0.50 / -0.48 | -0.24 / -0.66 | 0.175 | 17.125 |
| | Black-Male | 28 | -0.09 / -0.17 | -0.37 / -0.41 | 0.178 | 17.221 |
| | White-Female | 117 | +0.02 / +0.49 | +0.23 / +0.02 | 0.148 | 17.167 |
| | White-Male | 73 | +0.71 / +0.56 | +0.03 / +0.03 | 0.216 | 17.124 |
| Gemma-4-26B-A4B | Black-Female | 28 | +1.02 / +0.25 | -0.24 / -0.66 | 0.175 | 21.815 |
| | Black-Male | 28 | -0.03 / -0.04 | -0.37 / -0.41 | 0.178 | 21.158 |
| | White-Female | 117 | -0.28 / -0.63 | +0.23 / +0.02 | 0.148 | 21.274 |
| | White-Male | 73 | -0.64 / -0.59 | +0.03 / +0.03 | 0.216 | 20.672 |
| Gemma-4-31B | Black-Female | 28 | -0.34 / -0.31 | -0.24 / -0.66 | 0.175 | 25.888 |
| | Black-Male | 28 | +0.13 / +0.04 | -0.37 / -0.41 | 0.178 | 25.787 |
| | White-Female | 117 | +0.19 / +0.41 | +0.23 / +0.02 | 0.148 | 25.832 |
| | White-Male | 73 | +0.45 / +0.42 | +0.03 / +0.03 | 0.216 | 25.285 |
| Qwen3-14B | Black-Female | 28 | +0.53 / -0.11 | -0.24 / -0.66 | 0.175 | 1.817 |
| | Black-Male | 28 | +0.07 / -0.87 | -0.37 / -0.41 | 0.178 | 1.621 |
| | White-Female | 117 | -0.41 / +0.17 | +0.23 / +0.02 | 0.148 | 1.844 |
| | White-Male | 73 | -0.83 / -0.12 | +0.03 / +0.03 | 0.216 | 1.452 |
| Qwen3.6-27B | Black-Female | 28 | +0.15 / -0.61 | -0.24 / -0.66 | 0.175 | 5.420 |
| | Black-Male | 28 | +0.13 / -0.07 | -0.37 / -0.41 | 0.178 | 5.362 |
| | White-Female | 117 | -0.13 / -0.24 | +0.23 / +0.02 | 0.148 | 5.386 |
| | White-Male | 73 | +0.20 / +0.61 | +0.03 / +0.03 | 0.216 | 5.226 |
| Qwen3.6-35B-A3B | Black-Female | 28 | -0.08 / +0.07 | -0.24 / -0.66 | 0.175 | 3.112 |
| | Black-Male | 28 | -0.14 / +0.10 | -0.37 / -0.41 | 0.178 | 2.875 |
| | White-Female | 117 | +0.64 / +0.39 | +0.23 / +0.02 | 0.148 | 3.193 |
| | White-Male | 73 | +0.24 / +0.28 | +0.03 / +0.03 | 0.216 | 2.827 |

**Table 9: Warmth/competence bias by crossed race × gender group, model versus human, standardized.** The same four groups as Tab. S.11, collapsed across source study and standardized to $z$-scores as in Tab. 8. Raw, study-broken values are in Tab. S.11.

| Model | Grouping | Probe | $a$ | $b$ | IE | 95% CI |
|---|---|---|---|---|---|---|
| Gemma-3-12B | Race (Black=1) | Warmth | -0.502 | +0.171 | -0.086 | [-0.186, +0.019] |
| | Race (Black=1) | Competence | -0.501 | +0.174 | -0.087 | [-0.188, +0.018] |
| | Gender (Female=1) | Warmth | +0.072 | +0.236 | +0.017 | [-0.011, +0.056] |
| | Gender (Female=1) | Competence | +0.068 | +0.240 | +0.016 | [-0.012, +0.055] |
| Gemma-3-27B | Race (Black=1) | Warmth | -0.391 | -0.114 | +0.045 | [-0.008, +0.106] |
| | Race (Black=1) | Competence | -0.383 | -0.108 | +0.041 | [-0.010, +0.102] |
| | Gender (Female=1) | Warmth | +0.093 | -0.066 | -0.006 | [-0.027, +0.006] |
| | Gender (Female=1) | Competence | +0.096 | -0.055 | -0.005 | [-0.025, +0.007] |
| Llama-3.1-8B | Race (Black=1) | Warmth | +0.519 | +0.366 | $+0.190^{\dagger}$ | [+0.111, +0.292] |
| | Race (Black=1) | Competence | +0.218 | +0.185 | $+0.040^{\dagger}$ | [+0.004, +0.090] |
| | Gender (Female=1) | Warmth | +0.027 | -0.112 | -0.003 | [-0.022, +0.016] |
| | Gender (Female=1) | Competence | +0.208 | -0.012 | -0.003 | [-0.035, +0.030] |
| Gemma-4-12B | Race (Black=1) | Warmth | -0.260 | -0.109 | +0.028 | [-0.003, +0.069] |
| | Race (Black=1) | Competence | -0.515 | +0.291 | $-0.150^{\dagger}$ | [-0.295, -0.019] |
| | Gender (Female=1) | Warmth | -0.327 | -0.119 | +0.039 | [-0.004, +0.079] |
| | Gender (Female=1) | Competence | -0.032 | -0.024 | +0.001 | [-0.015, +0.011] |
| Gemma-4-26B-A4B | Race (Black=1) | Warmth | +0.476 | +0.173 | +0.082 | [-0.038, +0.194] |
| | Race (Black=1) | Competence | +0.387 | +0.339 | $+0.131^{\dagger}$ | [+0.060, +0.231] |
| | Gender (Female=1) | Warmth | +0.213 | +0.181 | $+0.038^{\dagger}$ | [+0.007, +0.078] |
| | Gender (Female=1) | Competence | -0.073 | +0.386 | -0.028 | [-0.076, +0.023] |
| Gemma-4-31B | Race (Black=1) | Warmth | -0.372 | +0.143 | -0.053 | [-0.164, +0.005] |
| | Race (Black=1) | Competence | -0.465 | +0.494 | $-0.230^{\dagger}$ | [-0.483, -0.049] |
| | Gender (Female=1) | Warmth | -0.094 | +0.088 | -0.008 | [-0.042, +0.002] |
| | Gender (Female=1) | Competence | +0.010 | +0.186 | +0.002 | [-0.029, +0.026] |
| Qwen3-14B | Race (Black=1) | Warmth | +0.488 | +0.165 | $+0.081^{\dagger}$ | [+0.013, +0.156] |
| | Race (Black=1) | Competence | -0.255 | +0.517 | $-0.132^{\dagger}$ | [-0.217, -0.058] |
| | Gender (Female=1) | Warmth | +0.210 | +0.014 | +0.003 | [-0.017, +0.024] |
| | Gender (Female=1) | Competence | +0.218 | +0.347 | $+0.076^{\dagger}$ | [+0.029, +0.141] |
| Qwen3.6-27B | Race (Black=1) | Warmth | +0.070 | +0.411 | +0.029 | [-0.029, +0.106] |
| | Race (Black=1) | Competence | -0.169 | +0.290 | $-0.049^{\dagger}$ | [-0.103, -0.011] |
| | Gender (Female=1) | Warmth | -0.021 | +0.415 | -0.009 | [-0.059, +0.047] |
| | Gender (Female=1) | Competence | -0.320 | +0.383 | $-0.123^{\dagger}$ | [-0.206, -0.061] |
| Qwen3.6-35B-A3B | Race (Black=1) | Warmth | -0.228 | +0.454 | $-0.103^{\dagger}$ | [-0.169, -0.046] |
| | Race (Black=1) | Competence | -0.097 | +0.354 | -0.034 | [-0.080, +0.006] |
| | Gender (Female=1) | Warmth | +0.249 | +0.257 | $+0.064^{\dagger}$ | [+0.027, +0.113] |
| | Gender (Female=1) | Competence | +0.124 | +0.234 | $+0.029^{\dagger}$ | [+0.001, +0.067] |

**Table 10: Bootstrap mediation of name-group callback disparities through warmth and competence probe scores, all nine models.** Path: name group → probe score → callback margin, standardized coefficients, Baron–Kenny/Preacher–Hayes bootstrap (5,000 resamples, seed 20260527, $n = 269$, race subset $n = 227$). $a$ is the coefficient of group on probe score; $b$ is the coefficient of probe score on callback margin, controlling for group; IE $= a \times b$ is the indirect (mediated) effect. $^{\dagger}$ marks a 95% bootstrap CI that excludes zero. 14 of 36 tests are significant at this uncorrected threshold; only the Llama-3.1-8B race–warmth path survives a Bonferroni correction across all 36 tests ($\alpha = 0.05/36$), so the remaining paths listed here should be read as suggestive rather than confirmed.

## Discussion and Conclusion

*Work in progress.*

**Limitations.** This study has several limitations, listed individually here so each concern can be weighed against its own evidence rather than folded into continuous prose:

- **Circularity in stimulus generation**: The concept story corpus was generated by a large language model, raising a concern about circularity in a study that probes language model representations. The generating model (Claude, Anthropic) and the probed models (Gemma, Qwen, Llama) come from different organizations and architectures, so the warmth/competence contrasts are not trivially reproduced by a shared generator. Even so, AI-generated stimuli could share stylistic properties that inflate probe accuracy across model families; replication with human-authored or template-based stimuli is desirable.
- **Limited topic coverage**: The corpus covers 50 topics. Broader topic coverage and larger per-condition samples would further test how far the directions generalize.
- **Moderate probe-to-human alignment**: Across all nine models (Tab. 7), warmth correlations with human ratings range from $\rho = -0.287$ to $+0.388$ and competence correlations range from $\rho = -0.058$ (n.s.) to $+0.466$. Competence tracks human perception more consistently than warmth, which turns negative or non-significant in four of the nine models; the models with the strongest warmth alignment (Gemma-3-12B, Gemma-3-27B) reach only moderate correlations even at their best. Taken together, the model's internal probes capture a component of human social perception rather than replicating it faithfully, and warmth is the less reliably captured of the two constructs. The subset of models with inverted warmth alignment is discussed separately below.
- **Inverted warmth construct in two models**: The warmth anti-alignment in Llama-3.1-8B ($\rho = -0.287$) and Qwen3-14B ($\rho = -0.178$) means that for these models the mean-difference direction captures a construct systematically inverted relative to the human warmth scale; results for these models should be interpreted with this inversion in mind.
- **A fixed probe-layer heuristic, not validated per model**: We set probe_layer_frac = 0.66 for every model based on the relative depth reported for emotion representations in Sofroniew et al. (2026), applying it as a static, prespecified choice rather than testing separability at that depth on each model individually. A full-depth layer sweep on the training story corpus shows that this heuristic layer rarely coincides with the layer of peak separability: at Gemma-3-12B the selected layer reaches a warmth Cohen's $d$ of 2.68, while the best-separated layer in the same sweep reaches 6.27; Gemma-3-27B (2.95 versus 5.10) and Gemma-4-31B (7.56 versus 11.49) show comparable gaps. This mismatch does not fully explain the weak or inverted name-level correlations reported above: Gemma-4-12B already reaches a high separability score at its selected layer ($d = 8.46$), yet the resulting warmth probe fails to correlate with human name ratings ($\rho = +0.009$, n.s.). Story-level separability, which guided layer selection, therefore does not guarantee that the resulting direction generalizes to bare applicant names, a case the training stories never present. Validating or tuning the probe layer per model, rather than transferring a single depth from a different study, is likely more principled; whether averaging or weighting probe projections across several separable layers would recover a stronger name-level signal than the single fixed layer is a question we leave open.
- **Ceiling effects in cross-validated accuracy**: 5-fold and topic-holdout accuracy reach 1.000 with zero variance across folds for every model and both axes. This is consistent with the large Cohen's $d$ values reported above (for example $d = 2.68$ for warmth at Gemma-3-12B), but a ceiling this uniform is also expected on statistical grounds alone: with only 100 stories per axis and residual-stream vectors of several thousand dimensions, near-perfect linear separability is likely regardless of whether the direction captures a meaningful concept. Topic-holdout accuracy was designed specifically to close the topic-vocabulary leakage route available to plain 5-fold splitting, described above, yet it saturates at the same ceiling; each topic-holdout fold still contains only around ten held-out topics against several-thousand-dimensional vectors, so the same high-dimension, low-sample-size

argument applies there as well, and the stricter split does not by itself rule it out. Cross-validated accuracy at ceiling therefore cannot distinguish between models or axes on its own, and should be read alongside the effect-size and split-half checks rather than as independent evidence of signal strength.

- **Fragile causal effect at scale**: The causal warmth steering effect at the hiring level is positive and strong at 12B but non-monotone and fragile at 27B, so the causal claim does not generalize across scale without qualification.
- **Callback-margin quantization**: Callback margins are quantized to a 0.125-unit grid due to bf16 precision in the model's output logits; this quantization is inherent to bf16 inference and is not resolved by float32 subtraction. We attempted a partial fix by casting the Yes/No logits to float32 before subtracting them, which removes rounding in the subtraction step itself, but the two operands are already bf16-quantized by the time they reach that subtraction, so the 0.125 grid persists regardless. A complete fix would require running inference itself in float32, which roughly doubles GPU memory usage and is likely infeasible for the 27B checkpoint on the available cluster hardware; we did not attempt it.
- **Narrow callback variance at some checkpoints**: At Gemma-3-12B the callback margin standard deviation is 0.14 with only seven unique values, making group-level disparity estimates at 12B unreliable; the 12B R4 results should not be cited as empirical findings. Llama-3.1-8B shows a comparably narrow spread (SD = 0.10, 10 unique values), so its group-level gaps should be read with the same caution. At Gemma-3-27B the larger variance (SD = 0.45, 19 unique values) makes group gaps of approximately 0.5 SD detectable, but the grid structure introduces discrete steps in all reported gap magnitudes. Qwen3-14B (SD = 0.36, 16 unique values) falls in the same usable range as 27B.
- **Uncorrected mediation tests**: Across all nine models, the thirty-six mediation tests are presented uncorrected for multiple comparisons; only the Llama-3.1-8B race–warmth indirect effect survives a Bonferroni-corrected threshold ($\alpha = 0.05/36$), and the smaller significant paths should be treated as suggestive rather than confirmed.
- **Simplified hiring prompt**: The hiring prompt is simplified relative to real-world screening contexts (single role, fixed qualifications), and results may differ under richer evaluation settings or different job types.
- **Open architectural question**: The architectural source of the elevated warmth/competence vector cosine observed in Gemma-3 relative to Qwen3 and Llama-3.1 across all network depths remains an open question.
- **Steering-strength range calibrated on two models**: We fixed the local steering range after testing saturation on Gemma-3-12B and Gemma-3-27B only, then applied it uniformly to all nine models; a more thorough methodology would calibrate this range separately for each model rather than reusing one pair's calibration across architectures that differ substantially in scale.
- **Raw vectors only in the disparity and mediation pipeline**: The probe-vs-human correlations, group disparities, and mediation tests all use the raw dense directions; no denoised counterpart is computed for this pipeline, unlike the concept-level and hiring-steering causal tests, both of which report a denoised robustness check. Whether the disparity and mediation results are robust to the same PCA-denoising procedure is left open.

**Future Work.** The immediate next step is expanding the hiring stimuli to additional job roles to test whether results generalise beyond administrative assistant. The SAE decomposition performed here on Gemma-3 can be extended to Llama-3.1 using the Llama Scope sparse autoencoders (He et al., 2024), enabling a model-family comparison at the feature level. Validating the concept probes against a human-authored warmth/competence dataset would address the circularity concern associated with AI-generated stimuli. On the social-science side, extending the hiring stimuli beyond names to category-level signals (age, disability, religion) would broaden the scope of the benchmark comparison and test whether the representational account generalises beyond race and gender. Translating these findings into practical debiasing interventions and evaluating their robustness to adversarial prompting is a natural next step for applied work.

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# Supplementary Materials

## Details on Methods

### Compute and Hardware

All experiments ran on two compute environments. The SCCKN cluster (Universität Konstanz) supplied NVIDIA RTX PRO 6000 Blackwell GPUs (96 GB VRAM), which ran the Gemma-3, Llama-3.1, and Qwen3 jobs (Gemma-3-12B, Gemma-3-27B, Llama-3.1-8B, Qwen3-14B) and most of the extraction and steering jobs for the Gemma-4 and Qwen3.6 checkpoints. Gemma-4-12B is a partial exception: its Stage 1 activations were first extracted on an NVIDIA L40, and a later retry ran on an L40S instead. A dedicated reproducibility audit on the exact L40 hardware class found that both device classes agree on which layer peaks for warmth and competence (layers 26 and 27, respectively), but differ non-negligibly at some layers (maximum competence Cohen's $d$ difference of 0.914, cosine difference of 0.095). The exact-L40 run, hardware-matched to the original Stage 1 extraction, is the one reported throughout the main text. Select calibrated-steering replications for the Gemma-4 and Qwen3.6 checkpoints (all three Gemma-4 models and both Qwen3.6 models) instead ran on NVIDIA H100 GPUs at the Collective Computational Unit (CCU). The authors acknowledge support by the local computing resources through the core facility SCCKN. The authors additionally acknowledge the Collective Computational Unit (CCU) for providing H100 GPU access used in part of this work.

### Story Corpus by Topic

**Table S.1:** Story corpus by topic ($n = 50$). Every topic contributes exactly one story to each of the four conditions (high warmth, low warmth, high competence, low competence), for 200 stories total, and every protagonist is nameless and referred to only as "they" throughout (see Story Preparation). The word-count range gives the shortest and longest of a topic's four stories. The Category column groups topics into the ten domains named in Story Preparation; these labels are our own assignment for readability and are not a field in the underlying stimuli data.

| No. | Topic | Category | Words |
|---|---|---|---|
| 1 | A team meeting where a decision needs to be made under time pressure | Workplace | 130–142 |
| 2 | Receiving critical feedback from a supervisor in front of colleagues | Workplace | 100–104 |
| 3 | Presenting results that did not go as expected to senior management | Workplace | 95–103 |
| 4 | Discovering that a colleague made a serious mistake before the deadline | Workplace | 99–103 |
| 5 | Mentoring a junior colleague on their first week at the job | Workplace | 97–105 |
| 6 | Handling an escalating complaint from a dissatisfied customer | Workplace | 93–100 |
| 7 | Negotiating project scope with a client who keeps changing requirements | Workplace | 91–92 |
| 8 | Organising a company event with a limited budget and unclear instructions | Workplace | 89–96 |
| 9 | Taking over a project mid-way when the previous lead leaves unexpectedly | Workplace | 90–101 |
| 10 | Responding to an emergency system failure at 2am on call | Workplace | 92–102 |
| 11 | Being asked to do something you believe is wrong by a manager | Learning | 91–102 |
| 12 | Learning a completely new technical skill under pressure and alone | Learning | 97–104 |
| 13 | Debugging a complex problem with no documentation and no internet | Learning | 94–99 |
| 14 | Preparing for a high-stakes exam with only two days left | Learning | 93–97 |
| 15 | Trying to understand a subject that feels overwhelming at first | Learning | 93–101 |
| 16 | Explaining a technical concept to someone with no background in it | Learning | 126–144 |
| 17 | Helping a stranger who dropped their groceries in the street | Social | 88–97 |
| 18 | A difficult conversation with a family member about money | Social | 90–100 |
| 19 | Being the only person at a party who does not know anyone | Social | 92–95 |
| 20 | Comforting a friend who just received bad news | Social | 89–102 |
| 21 | Waiting for important medical test results | Health | 92–109 |
| 22 | Supporting a parent who is beginning to lose their independence | Health | 90–95 |
| 23 | Managing a chronic illness while trying to maintain a normal schedule | Health | 88–96 |
| 24 | Recovering from an injury when you were counting on being active | Health | 90–105 |
| 25 | Organising a neighbourhood clean-up with volunteers who cancel last minute | Community | 129–140 |
| 26 | Attending a community meeting where people strongly disagree | Community | 91–100 |

*continued on next page*

| No. | Topic | Category | Words |
|---|---|---|---|
| 27 | Volunteering at a food bank on an unexpectedly chaotic shift | Community | 89–99 |
| 28 | Running for a small local position against an experienced opponent | Community | 93–106 |
| 29 | Advocating for a change at the local school as a parent | Community | 90–101 |
| 30 | Deciding whether to take a pay cut for a job you believe in | Finance | 88–95 |
| 31 | Realising mid-month that you have far less money than expected | Finance | 90–104 |
| 32 | Negotiating a salary for the first time in a new job offer | Finance | 90–103 |
| 33 | Dealing with a billing dispute that has lasted several months | Finance | 89–103 |
| 34 | Coaching a sports team through a long losing streak | Sport & Creative | 129–143 |
| 35 | Performing music in public for the first time and making a mistake | Sport & Creative | 90–101 |
| 36 | Finishing a creative project that has been stalled for months | Sport & Creative | 97–107 |
| 37 | Being a referee in a game where the players are friends | Sport & Creative | 93–98 |
| 38 | Entering a competition knowing you are not the favourite | Sport & Creative | 91–107 |
| 39 | Missing a connecting flight with no phone battery and no local currency | Travel | 125–142 |
| 40 | Navigating a city you have never been to, in a country whose language you do not speak | Travel | 90–100 |
| 41 | Being stranded overnight due to a transport strike with strangers | Travel | 93–107 |
| 42 | Starting over in a new country with no local contacts | Travel | 91–99 |
| 43 | Arriving at accommodation that is nothing like what was advertised | Travel | 93–102 |
| 44 | Realising your important data was lost and there is no backup | Technology | 90–95 |
| 45 | Trying to set up a complex technical system with no instructions | Technology | 95–105 |
| 46 | Being asked to fix someone else's code under a tight deadline | Technology | 91–100 |
| 47 | Teaching an elderly relative how to use a new device they are afraid of | Technology | 90–102 |
| 48 | Apologising sincerely to someone you have genuinely hurt | Personal Milestones | 94–104 |
| 49 | Being the first person in your family to pursue a university degree | Personal Milestones | 92–102 |
| 50 | Caring for a newborn alone for the first time with no experience | Personal Milestones | 90–98 |

## Story Generation and System Prompt

The corpus was written manually, condition by condition, by Claude Opus 4.8 against a fixed specification: a single nameless protagonist referred to only as "they," a per-condition list of forbidden words that would name the concept directly, a show-don't-tell instruction to convey warmth or competence through actions rather than labels, and the 90–150 word length band enforced across every condition. Tab. S.2 reserves the exact generation and system prompt text.

| |
|---|
| Work in progress |

**Table S.2:** Generation and system prompt supplied to the writing model. This table is a placeholder pending the verbatim prompt text.

## Example Concept Stories

Tab. S.3 quotes one story from each of the four conditions, all four sharing a single topic, so that the contrast between conditions is visible independent of the situation described.

**Table S.3:** Example stories, one per condition, sharing a single topic (topic 1 of Tab. S.1). The complete set of all 200 story texts is available at `[WORK IN PROGRESS]`.

| Condition | Story (topic: "A team meeting where a decision needs to be made under time pressure") |
|---|---|
| High warmth | Noticing that one teammate hadn't spoken since the start, they paused and asked what that person was seeing from their side of the project. When two colleagues talked over each other, they waited, then restated each point so neither felt brushed aside. Aware that everyone had skipped lunch, they pushed the open box of crackers to the middle of the table. As the deadline pressed in, they made sure the quietest intern's worry about the rollout went up on the board alongside everyone else's. 'We decide together or we don't decide well,' they said, watching faces rather than the clock. Once the group chose a direction, they stayed behind to walk the teammate who'd looked lost through what had been agreed, and afterwards messaged the intern to say the question raised had changed the final plan. |

*continued on next page*

| Condition | Story (topic: "A team meeting where a decision needs to be made under time pressure") |
|---|---|
| Low warmth | Opening the meeting, they announced they already knew the right call and that the discussion was a formality. When a colleague tried to raise a risk with the vendor, they kept scrolling their phone and said there was no time for hypotheticals. An intern asked a clarifying question; they answered someone else, as though it hadn't been spoken. They read the group's hesitation as wasted time rather than doubt and pushed for a vote. 'I don't need everyone comfortable, I need a decision,' they said, not looking up. When a teammate mentioned being away during the rollout and feeling worried, they replied that it wasn't their problem to manage. The group agreed quickly, mostly to end the meeting, and they left first without waiting for anyone to gather their things. |
| High competence | Having read the vendor report the night before, they broke the stall by putting three options on the board, each already filled in with its cost and failure risk. They flagged that the cheapest option hid a renewal clause that would double the price in year two, buried on page forty. When someone proposed a fourth path, they tested it aloud against the deadline and showed exactly where it broke. They set a ten-minute limit per section and tracked it, steering the group back whenever it drifted. Before the vote they summarised the trade-offs in two sentences so no one chose blind. Once the decision was made, they assigned every follow-up to a named owner with a date, typed it into the shared doc while the group watched, and read it back. The meeting finished four minutes early, every action already logged. |
| Low competence | They started the meeting before locating the file, then spent minutes searching their inbox while the room waited. The numbers they finally opened were from the previous quarter, though they insisted these were current until someone pointed at the date. They proposed the option with the highest upside without mentioning its cost, and when asked about the budget guessed a figure that proved wrong by half. Each risk that came up went onto a sticky note they then misplaced. They changed the agenda twice mid-discussion, reopening debates already settled. With ten minutes left they realised the one person who owned the rollout had never been invited. The group agreed to revisit it the next day, having decided nothing, and they promised a summary that by morning contradicted what most people remembered agreeing. |

## PCA Denoising, All Nine Models

The warmth and competence direction vectors share a substantial common component that reflects general narrative valence: stories that portray a protagonist positively (whether warmly or competently) occupy a similar region of the residual stream, and the same holds for negatively valenced stories. To separate conceptual content from this shared evaluative polarity, we follow the neutral-corpus denoising procedure of Sofroniew et al. (2026). We build a neutral corpus of 1,500 Wikipedia introduction paragraphs, length-matched to the concept stories (90–200 words) and filtered to exclude emotionally charged or violent content, and extract residual-stream activations at the same layer and pooling rule used for the concept stories. We then fit principal component analysis (PCA) on these neutral activations and project out, from both direction vectors, the top components that together explain at least 50 percent of neutral variance. For Gemma-3-12B, the first principal component alone explains 56.1 percent of neutral variance, and removing it lowers the cosine between the warmth and competence vectors from 0.749 to 0.530. For Gemma-3-27B, 43 components are needed to cross the 50 percent threshold (50.2 percent explained), lowering the inter-axis cosine from 0.708 to 0.487. The remaining post-denoising cosine, 0.53 and 0.49 respectively, is close to the genuine warmth-competence correlation observed in human ratings ($r \approx 0.61$; Fiske et al. 2002), which suggests that the residual overlap reflects real conceptual co-variation rather than a failure of the denoising step. The denoised vectors serve as a robustness check; the raw dense directions define the primary cross-model causal comparisons and the callback-transition census reported in the main text. Tab. S.4 extends this same procedure to all nine models.

| Model | $k$ (PCs removed) | Variance kept | cos(W,C) before | cos(W,C) after |
|---|---|---|---|---|
| Gemma-3-12B | 1 | 56.1% | 0.749 | 0.530 |
| Gemma-3-27B | 43 | 50.2% | 0.708 | 0.487 |
| Llama-3.1-8B | 48 | 50.3% | 0.505 | 0.492 |
| Gemma-4-12B | 11 | 51.2% | 0.494 | 0.473 |
| Gemma-4-26B-A4B | 11 | 50.3% | 0.587 | 0.564 |
| Gemma-4-31B | 17 | 50.6% | 0.526 | 0.469 |
| Qwen3-14B | 19 | 50.5% | 0.536 | 0.510 |
| Qwen3.6-27B | 27 | 50.2% | 0.580 | 0.560 |
| Qwen3.6-35B-A3B | 17 | 50.3% | 0.619 | 0.595 |

**Table S.4: Neutral-corpus PCA denoising, all nine models.** $k$ is the number of leading principal components of 1,500 neutral Wikipedia activations removed from both direction vectors, the smallest $k$ crossing the 50% neutral-variance threshold. cos(W,C) before/after is the cosine between the warmth and competence directions, measured before and after removing those components. Denoising reduces cos(W,C) in every model but never eliminates it.

## Cross-Model Story-Ranking Agreement, All 36 Pairs

Tab. S.5 lists the Spearman correlation, both overall and within-condition, for every one of the $\binom{9}{2} = 36$ possible model pairs on both axes, the full table behind Fig. 4 in Results.

**Table S.5: Cross-model Spearman story-ranking agreement, all 36 model pairs.** For each pair and axis, every one of the 200 concept stories is projected onto both models' own warmth or competence direction, and the two resulting score lists are correlated with Spearman's $\rho$. "Overall" includes the high/low condition separation itself; "within-condition" restricts the correlation to stories sharing one condition, so it reflects agreement on finer-grained story ordering rather than the coarse high-versus-low split. The four per-condition $\rho$ values behind the within-condition summary are in `results/tables/cross_model_agreement_9model.csv`.

| Model A | Model B | Overall $\rho$ | Within-condition $\rho$ |
|---|---|---|---|
| *Warmth* | | | |
| Gemma-3-12B | Gemma-3-27B | 0.818 | 0.555 |
| Gemma-3-12B | Llama-3.1-8B | 0.768 | 0.203 |
| Gemma-3-12B | Gemma-4-12B | 0.766 | 0.195 |
| Gemma-3-12B | Gemma-4-26B-A4B | 0.741 | 0.095 |
| Gemma-3-12B | Gemma-4-31B | 0.801 | 0.285 |
| Gemma-3-12B | Qwen3-14B | 0.760 | 0.160 |
| Gemma-3-12B | Qwen3.6-27B | 0.803 | 0.308 |
| Gemma-3-12B | Qwen3.6-35B-A3B | 0.797 | 0.251 |
| Gemma-3-27B | Llama-3.1-8B | 0.783 | 0.208 |
| Gemma-3-27B | Gemma-4-12B | 0.757 | 0.120 |
| Gemma-3-27B | Gemma-4-26B-A4B | 0.759 | 0.124 |
| Gemma-3-27B | Gemma-4-31B | 0.762 | 0.182 |
| Gemma-3-27B | Qwen3-14B | 0.777 | 0.223 |
| Gemma-3-27B | Qwen3.6-27B | 0.792 | 0.308 |
| Gemma-3-27B | Qwen3.6-35B-A3B | 0.760 | 0.174 |
| Llama-3.1-8B | Qwen3-14B | 0.978 | 0.833 |
| Llama-3.1-8B | Qwen3.6-27B | 0.945 | 0.641 |
| Llama-3.1-8B | Qwen3.6-35B-A3B | 0.896 | 0.541 |
| Gemma-4-12B | Llama-3.1-8B | 0.931 | 0.476 |
| Gemma-4-12B | Gemma-4-26B-A4B | 0.911 | 0.434 |
| Gemma-4-12B | Gemma-4-31B | 0.940 | 0.574 |
| Gemma-4-12B | Qwen3-14B | 0.941 | 0.516 |
| Gemma-4-12B | Qwen3.6-27B | 0.920 | 0.453 |
| Gemma-4-12B | Qwen3.6-35B-A3B | 0.884 | 0.451 |
| Gemma-4-26B-A4B | Llama-3.1-8B | 0.932 | 0.586 |
| Gemma-4-26B-A4B | Gemma-4-31B | 0.905 | 0.467 |
| Gemma-4-26B-A4B | Qwen3-14B | 0.948 | 0.698 |
| Gemma-4-26B-A4B | Qwen3.6-27B | 0.919 | 0.530 |
| Gemma-4-26B-A4B | Qwen3.6-35B-A3B | 0.892 | 0.488 |

*Tab. S.5 continued*

| Model A | Model B | Overall $\rho$ | Within-condition $\rho$ |
|---|---|---|---|
| Gemma-4-31B | Llama-3.1-8B | 0.908 | 0.428 |
| Gemma-4-31B | Qwen3-14B | 0.922 | 0.497 |
| Gemma-4-31B | Qwen3.6-27B | 0.926 | 0.502 |
| Gemma-4-31B | Qwen3.6-35B-A3B | 0.902 | 0.479 |
| Qwen3-14B | Qwen3.6-27B | 0.953 | 0.724 |
| Qwen3-14B | Qwen3.6-35B-A3B | 0.904 | 0.595 |
| Qwen3.6-27B | Qwen3.6-35B-A3B | 0.930 | 0.685 |
| *Competence* | | | |
| Gemma-3-12B | Gemma-3-27B | 0.825 | 0.557 |
| Gemma-3-12B | Llama-3.1-8B | 0.782 | 0.295 |
| Gemma-3-12B | Gemma-4-12B | 0.786 | 0.326 |
| Gemma-3-12B | Gemma-4-26B-A4B | 0.792 | 0.307 |
| Gemma-3-12B | Gemma-4-31B | 0.812 | 0.413 |
| Gemma-3-12B | Qwen3-14B | 0.795 | 0.364 |
| Gemma-3-12B | Qwen3.6-27B | 0.799 | 0.330 |
| Gemma-3-12B | Qwen3.6-35B-A3B | 0.785 | 0.282 |
| Gemma-3-27B | Llama-3.1-8B | 0.834 | 0.304 |
| Gemma-3-27B | Gemma-4-12B | 0.831 | 0.284 |
| Gemma-3-27B | Gemma-4-26B-A4B | 0.823 | 0.258 |
| Gemma-3-27B | Gemma-4-31B | 0.819 | 0.311 |
| Gemma-3-27B | Qwen3-14B | 0.836 | 0.314 |
| Gemma-3-27B | Qwen3.6-27B | 0.829 | 0.262 |
| Gemma-3-27B | Qwen3.6-35B-A3B | 0.800 | 0.204 |
| Llama-3.1-8B | Qwen3-14B | 0.992 | 0.899 |
| Llama-3.1-8B | Qwen3.6-27B | 0.979 | 0.757 |
| Llama-3.1-8B | Qwen3.6-35B-A3B | 0.943 | 0.513 |
| Gemma-4-12B | Llama-3.1-8B | 0.964 | 0.590 |
| Gemma-4-12B | Gemma-4-26B-A4B | 0.960 | 0.618 |
| Gemma-4-12B | Gemma-4-31B | 0.952 | 0.645 |
| Gemma-4-12B | Qwen3-14B | 0.970 | 0.646 |
| Gemma-4-12B | Qwen3.6-27B | 0.965 | 0.621 |
| Gemma-4-12B | Qwen3.6-35B-A3B | 0.935 | 0.476 |
| Gemma-4-26B-A4B | Llama-3.1-8B | 0.974 | 0.751 |
| Gemma-4-26B-A4B | Gemma-4-31B | 0.947 | 0.619 |
| Gemma-4-26B-A4B | Qwen3-14B | 0.980 | 0.802 |
| Gemma-4-26B-A4B | Qwen3.6-27B | 0.972 | 0.727 |
| Gemma-4-26B-A4B | Qwen3.6-35B-A3B | 0.945 | 0.532 |
| Gemma-4-31B | Llama-3.1-8B | 0.943 | 0.577 |
| Gemma-4-31B | Qwen3-14B | 0.952 | 0.663 |
| Gemma-4-31B | Qwen3.6-27B | 0.952 | 0.633 |
| Gemma-4-31B | Qwen3.6-35B-A3B | 0.928 | 0.515 |
| Qwen3-14B | Qwen3.6-27B | 0.981 | 0.783 |
| Qwen3-14B | Qwen3.6-35B-A3B | 0.952 | 0.584 |
| Qwen3.6-27B | Qwen3.6-35B-A3B | 0.957 | 0.630 |

## Gemma Scope Steering Robustness, Gemma-3-12B/27B

Three further checks probe the direction-specificity result reported in Results. Tab. S.6 first asks whether the Gemma Scope 2 decomposition itself is trustworthy: reconstruction cosine stays above 0.995 at every SAE width for both models, so the sparse representation preserves the original activation well, even though topic-holdout accuracy inside that sparse space and the decoded direction's alignment with the dense direction both vary more across widths.

| Model | Width | Recon. cosine | Active feats. | Topic-CV (W/C) | Decoded cosine (W/C) |
|---|---|---|---|---|---|
| Gemma-3-12B | 16k | 0.997 | 8.6 | 0.94 / 0.78 | 0.73 / 0.72 |
| Gemma-3-12B | 65k | 0.997 | 6.3 | 0.72 / 0.77 | 0.74 / 0.64 |
| Gemma-3-12B | 262k | 0.998 | 5.6 | 0.70 / 0.69 | 0.75 / 0.72 |
| Gemma-3-27B | 16k | 0.995 | 7.6 | 0.84 / 0.76 | 0.61 / 0.40 |
| Gemma-3-27B | 65k | 0.995 | 6.9 | 0.77 / 0.75 | 0.58 / 0.57 |
| Gemma-3-27B | 262k | 0.995 | 5.3 | 0.71 / 0.79 | 0.65 / 0.63 |

**Table S.6: Gemma Scope 2 SAE reconstruction quality.** Recon. cosine is the cosine between the original activation and its SAE reconstruction; active features is the mean number of nonzero SAE features per story; topic-CV is 5-fold topic-holdout accuracy for warmth/competence classification inside the sparse feature space; decoded cosine is the alignment between the decoded SAE direction and the raw dense direction.

Tab. S.7 removes, rather than adds, a feature group and tracks the resulting change in the high-low margin gap.

| Model | Axis | Target axis | Shared | Opposing axis | Random |
|---|---|---|---|---|---|
| Gemma-3-12B | Warmth | +0.225 | +0.287 | +0.100 | +0.013 |
| Gemma-3-12B | Competence | -0.050 | +0.113 | +0.125 | +0.000 |
| Gemma-3-27B | Warmth | +0.075 | -0.725 | +0.044 | +0.200 |
| Gemma-3-27B | Competence | -0.087 | -0.319 | +0.013 | +0.025 |

**Table S.7: Feature ablation, Gemma-3-12B/27B.** Change in the high-versus-low margin gap when the named group of SAE features is zeroed out rather than added to; a negative value means ablation shrinks the gap. Shared-feature ablation shrinks the gap more than target-axis ablation in three of the four rows, echoing the substantial shared component already documented in PCA denoising rather than a clean axis-specific routing.

Tab. S.8 asks whether the two Gemma-3 scales converge on similar sparse features in the first place, matching each Gemma-3-12B feature to its closest Gemma-3-27B counterpart and comparing their story-level response profiles against a permutation null.

| Axis | $n$ matches | Obs. mean | Obs. median | Null mean [95% CI] | $p$ |
|---|---|---|---|---|---|
| Warmth | 135 | 0.644 | 0.663 | 0.354 [0.324, 0.387] | 0.0020 |
| Competence | 192 | 0.655 | 0.694 | 0.403 [0.365, 0.439] | 0.0020 |

**Table S.8: Cross-scale SAE feature matching, Gemma-3-12B $\leftrightarrow$ Gemma-3-27B.** For each matched feature pair, story-profile correlation is compared against a 500-permutation null. Observed mean/median well above the null mean, with a small permutation $p$-value, indicates the two scales share features with correlated story-level responses beyond chance.

# Additional Results

## Complete Positive-Steering Callback Transitions

Tab. S.9 reports every model-axis condition used in the positive-endpoint comparison. The table is the complete result; the matched panels in Fig. 7 are illustrative examples selected under a common family, application set, and steering strength. Gemma-3-12B, Gemma-3-27B, and the Qwen3.6 and Gemma-4 checkpoints use $\alpha = +0.10$. Llama-3.1-8B and Qwen3-14B use their available broad-regime endpoint of $\alpha = +0.50$, so raw effect sizes should not be compared directly across all rows.

Positive steering crosses the mean decision boundary in both directions among the Gemma-3 models and from No to Yes for Llama-3.1-8B. The other six models remain in the Yes category for every name, even when the mean margin decreases. This pattern reflects two distinct mechanisms. First, a concept direction constructed from high-minus-low stories is not constrained to align positively with the callback readout. Second, models beginning far from zero can move substantially without changing the categorical recommendation. The Gemma-3-27B response adds a third complication because its local dose-response reverses between +0.05 and +0.10, indicating a narrow nonlinear intervention regime.

| | | | Warmth | | | Competence | | |
|---|---|---|---|---|---|---|---|---|
| Model | $\alpha$ | Baseline | Endpoint | $\Delta$ margin | Transitions | Endpoint | $\Delta$ margin | Transitions |
| Gemma-3-12B | +0.10 | -0.194 (N) | +2.175 (Y) | +2.369 | 54 N→Y; 6 T→Y | +0.037 (Y) | +0.231 | 19 N→N, 13 N→T, 22 N→Y 1 T→N, 2 T→T, 3 T→Y |
| Gemma-3-27B | +0.10 | +1.169 (Y) | -1.490 (N) | -2.658 | 60 Y→N | -1.527 (N) | -2.696 | 60 Y→N |
| Llama-3.1-8B | +0.50 | -2.098 (N) | +1.073 (Y) | +3.171 | 60 N→Y | +0.072 (Y) | +2.170 | 15 N→N; 10 N→T; 35 N→Y |
| Gemma-4-12B | +0.10 | +17.188 (Y) | +18.237 (Y) | +1.049 | 60 Y→Y | +18.715 (Y) | +1.527 | 60 Y→Y |
| Gemma-4-26B-A4B | +0.10 | +21.245 (Y) | +21.307 (Y) | +0.062 | 60 Y→Y | +20.865 (Y) | -0.380 | 60 Y→Y |
| Gemma-4-31B | +0.10 | +25.615 (Y) | +25.173 (Y) | -0.442 | 60 Y→Y | +24.801 (Y) | -0.814 | 60 Y→Y |
| Qwen3-14B | +0.50 | +1.704 (Y) | +2.304 (Y) | +0.600 | 60 Y→Y | +1.321 (Y) | -0.383 | 60 Y→Y |
| Qwen3.6-27B | +0.10 | +5.346 (Y) | +6.542 (Y) | +1.196 | 60 Y→Y | +5.879 (Y) | +0.533 | 60 Y→Y |
| Qwen3.6-35B-A3B | +0.10 | +3.031 (Y) | +3.998 (Y) | +0.967 | 60 Y→Y | +3.465 (Y) | +0.433 | 60 Y→Y |

**Table S.9: Complete positive-endpoint callback transitions.** Mean baseline and steered callback margins are followed by their decision category in parentheses. Transition cells list every nonzero individual-name transition among No (N), Tie (T), and Yes (Y); each cell sums to 60.

## Name-Level Warmth/Competence by Marginal Group, Raw Values by Study

Tab. S.10 is the raw-value companion to Tab. 8 in the main text: the same race and gender marginal groups, broken down by source study rather than blended across it, since a name rated under more than one correspondence study can carry a substantially different callback rate across studies (Bertrand's "aisha" callback is 0.022, Kline's is 0.246). Model warmth and competence values are raw, unnormalized projections onto the concept direction, and human warmth and competence are the 0-100 Likert averages behind them, so neither is comparable in magnitude across models or against each other; the standardized ($z$-score) main-text version is built for that comparison.

**Table S.10: Warmth/competence bias by marginal demographic group, broken down by source study, raw values.** Companion to Tab. 8: the same race and gender marginal groups, not blended across studies. Model warmth/competence are raw (unnormalized) projections and are not comparable in magnitude across models or against the 0-100 human warmth/competence scale; see Tab. 8 for the standardized ($z$-score) comparison.

| Model | Axis | Group × Study | $n$ | Model warmth/competence | Human warmth/competence | Human callback |
|---|---|---|---|---|---|---|
| Gemma-3-12B | Race | Black / Bertrand | 18 | 37751.85 / 40473.48 | 57.96 / 60.93 | 0.062 |
| | | Black / Kline | 38 | 37704.47 / 40424.58 | 46.37 / 45.61 | 0.231 |
| | | White / Bertrand | 18 | 38512.67 / 41287.53 | 55.30 / 61.95 | 0.094 |
| | | White / Kline | 38 | 38533.82 / 41314.30 | 62.26 / 62.77 | 0.251 |
| | | White / Farber | 12 | 38682.20 / 41473.86 | 58.41 / 60.51 | 0.111 |
| | | White / Neumark | 122 | 38404.78 / 41176.39 | 53.69 / 55.70 | 0.168 |
| | | Female / Bertrand | 18 | 38142.21 / 40888.71 | 60.75 / 62.87 | 0.083 |
| | | Female / Kline | 38 | 38091.11 / 40834.26 | 55.49 / 52.14 | 0.241 |
| | | Female / Farber | 12 | 38682.20 / 41473.86 | 58.41 / 60.51 | 0.111 |
| | | Female / Neumark | 77 | 38474.98 / 41250.92 | 54.16 / 55.93 | 0.132 |
| | | Male / Bertrand | 18 | 38122.30 / 40872.30 | 52.50 / 60.01 | 0.074 |
| | | Male / Kline | 38 | 38147.17 / 40904.62 | 53.13 / 56.24 | 0.240 |
| | | Male / Neumark | 45 | 38284.64 / 41048.87 | 52.88 / 55.30 | 0.229 |
| Gemma-3-27B | Race | Black / Bertrand | 18 | 26190.79 / 25237.29 | 57.96 / 60.93 | 0.062 |
| | | Black / Kline | 38 | 26131.12 / 25178.45 | 46.37 / 45.61 | 0.231 |
| | | White / Bertrand | 18 | 26754.42 / 25759.67 | 55.30 / 61.95 | 0.094 |
| | | White / Kline | 38 | 26735.26 / 25740.82 | 62.26 / 62.77 | 0.251 |
| | | White / Farber | 12 | 26786.33 / 25790.22 | 58.41 / 60.51 | 0.111 |
| | | White / Neumark | 122 | 26556.63 / 25570.14 | 53.69 / 55.70 | 0.168 |
| | | Female / Bertrand | 18 | 26476.82 / 25508.09 | 60.75 / 62.87 | 0.083 |
| | | Female / Kline | 38 | 26407.84 / 25437.11 | 55.49 / 52.14 | 0.241 |
| | | Female / Farber | 12 | 26786.33 / 25790.22 | 58.41 / 60.51 | 0.111 |
| | | Female / Neumark | 77 | 26626.71 / 25638.55 | 54.16 / 55.93 | 0.132 |
| | | Male / Bertrand | 18 | 26468.39 / 25488.87 | 52.50 / 60.01 | 0.074 |
| | | Male / Kline | 38 | 26458.54 / 25482.15 | 53.13 / 56.24 | 0.240 |
| | | Male / Neumark | 45 | 26436.70 / 25453.09 | 52.88 / 55.30 | 0.229 |
| Llama-3.1-8B | Race | Black / Bertrand | 18 | -1.27 / -0.16 | 57.96 / 60.93 | 0.062 |
| | | Black / Kline | 38 | -1.26 / -0.16 | 46.37 / 45.61 | 0.231 |
| | | White / Bertrand | 18 | -1.35 / -0.18 | 55.30 / 61.95 | 0.094 |
| | | White / Kline | 38 | -1.35 / -0.18 | 62.26 / 62.77 | 0.251 |
| | | White / Farber | 12 | -1.34 / -0.17 | 58.41 / 60.51 | 0.111 |
| | | White / Neumark | 122 | -1.34 / -0.17 | 53.69 / 55.70 | 0.168 |
| | | Female / Bertrand | 18 | -1.30 / -0.16 | 60.75 / 62.87 | 0.083 |
| | | Female / Kline | 38 | -1.30 / -0.16 | 55.49 / 52.14 | 0.241 |
| | | Female / Farber | 12 | -1.34 / -0.17 | 58.41 / 60.51 | 0.111 |
| | | Female / Neumark | 77 | -1.33 / -0.17 | 54.16 / 55.93 | 0.132 |
| | | Male / Bertrand | 18 | -1.31 / -0.18 | 52.50 / 60.01 | 0.074 |
| | | Male / Kline | 38 | -1.31 / -0.17 | 53.13 / 56.24 | 0.240 |
| | | Male / Neumark | 45 | -1.34 / -0.17 | 52.88 / 55.30 | 0.229 |

*Tab. S.10 continued*

| Model | Axis | Group × Study | $n$ | Model warmth/competence | Human warmth/competence | Human callback |
|---|---|---|---|---|---|---|
| Gemma-4-12B | Race | Black / Bertrand | 18 | 12.20 / 18.46 | 57.96 / 60.93 | 0.062 |
| | | Black / Kline | 38 | 12.24 / 18.52 | 46.37 / 45.61 | 0.231 |
| | | White / Bertrand | 18 | 12.31 / 19.06 | 55.30 / 61.95 | 0.094 |
| | | White / Kline | 38 | 12.30 / 19.08 | 62.26 / 62.77 | 0.251 |
| | | White / Farber | 12 | 12.26 / 19.06 | 58.41 / 60.51 | 0.111 |
| | | White / Neumark | 122 | 12.29 / 18.98 | 53.69 / 55.70 | 0.168 |
| | | Female / Bertrand | 18 | 12.25 / 18.73 | 60.75 / 62.87 | 0.083 |
| | | Female / Kline | 38 | 12.25 / 18.74 | 55.49 / 52.14 | 0.241 |
| | | Female / Farber | 12 | 12.26 / 19.06 | 58.41 / 60.51 | 0.111 |
| | | Female / Neumark | 77 | 12.25 / 18.96 | 54.16 / 55.93 | 0.132 |
| | | Male / Bertrand | 18 | 12.27 / 18.79 | 52.50 / 60.01 | 0.074 |
| | | Male / Kline | 38 | 12.29 / 18.86 | 53.13 / 56.24 | 0.240 |
| | | Male / Neumark | 45 | 12.36 / 19.01 | 52.88 / 55.30 | 0.229 |
| Gemma-4-26B-A4B | Race | Black / Bertrand | 18 | 1.06 / 5.53 | 57.96 / 60.93 | 0.062 |
| | | Black / Kline | 38 | 1.07 / 5.56 | 46.37 / 45.61 | 0.231 |
| | | White / Bertrand | 18 | 0.88 / 5.44 | 55.30 / 61.95 | 0.094 |
| | | White / Kline | 38 | 0.87 / 5.43 | 62.26 / 62.77 | 0.251 |
| | | White / Farber | 12 | 0.88 / 5.42 | 58.41 / 60.51 | 0.111 |
| | | White / Neumark | 122 | 0.90 / 5.44 | 53.69 / 55.70 | 0.168 |
| | | Female / Bertrand | 18 | 1.03 / 5.48 | 60.75 / 62.87 | 0.083 |
| | | Female / Kline | 38 | 1.05 / 5.49 | 55.49 / 52.14 | 0.241 |
| | | Female / Farber | 12 | 0.88 / 5.42 | 58.41 / 60.51 | 0.111 |
| | | Female / Neumark | 77 | 0.92 / 5.44 | 54.16 / 55.93 | 0.132 |
| | | Male / Bertrand | 18 | 0.91 / 5.48 | 52.50 / 60.01 | 0.074 |
| | | Male / Kline | 38 | 0.89 / 5.49 | 53.13 / 56.24 | 0.240 |
| | | Male / Neumark | 45 | 0.86 / 5.43 | 52.88 / 55.30 | 0.229 |
| Gemma-4-31B | Race | Black / Bertrand | 18 | 45.30 / 78.18 | 57.96 / 60.93 | 0.062 |
| | | Black / Kline | 38 | 45.33 / 78.21 | 46.37 / 45.61 | 0.231 |
| | | White / Bertrand | 18 | 45.59 / 79.01 | 55.30 / 61.95 | 0.094 |
| | | White / Kline | 38 | 45.56 / 78.96 | 62.26 / 62.77 | 0.251 |
| | | White / Farber | 12 | 45.48 / 78.87 | 58.41 / 60.51 | 0.111 |
| | | White / Neumark | 122 | 45.49 / 78.75 | 53.69 / 55.70 | 0.168 |
| | | Female / Bertrand | 18 | 45.38 / 78.55 | 60.75 / 62.87 | 0.083 |
| | | Female / Kline | 38 | 45.36 / 78.47 | 55.49 / 52.14 | 0.241 |
| | | Female / Farber | 12 | 45.48 / 78.87 | 58.41 / 60.51 | 0.111 |
| | | Female / Neumark | 77 | 45.44 / 78.75 | 54.16 / 55.93 | 0.132 |
| | | Male / Bertrand | 18 | 45.51 / 78.64 | 52.50 / 60.01 | 0.074 |
| | | Male / Kline | 38 | 45.53 / 78.70 | 53.13 / 56.24 | 0.240 |
| | | Male / Neumark | 45 | 45.58 / 78.76 | 52.88 / 55.30 | 0.229 |
| Qwen3-14B | Race | Black / Bertrand | 18 | -8.48 / 17.36 | 57.96 / 60.93 | 0.062 |
| | | Black / Kline | 38 | -8.52 / 17.31 | 46.37 / 45.61 | 0.231 |
| | | White / Bertrand | 18 | -9.29 / 17.76 | 55.30 / 61.95 | 0.094 |
| | | White / Kline | 38 | -9.30 / 17.75 | 62.26 / 62.77 | 0.251 |
| | | White / Farber | 12 | -9.21 / 17.72 | 58.41 / 60.51 | 0.111 |
| | | White / Neumark | 122 | -9.20 / 17.63 | 53.69 / 55.70 | 0.168 |
| | | Female / Bertrand | 18 | -8.75 / 17.77 | 60.75 / 62.87 | 0.083 |
| | | Female / Kline | 38 | -8.70 / 17.68 | 55.49 / 52.14 | 0.241 |
| | | Female / Farber | 12 | -9.21 / 17.72 | 58.41 / 60.51 | 0.111 |
| | | Female / Neumark | 77 | -9.07 / 17.71 | 54.16 / 55.93 | 0.132 |
| | | Male / Bertrand | 18 | -9.02 / 17.35 | 52.50 / 60.01 | 0.074 |
| | | Male / Kline | 38 | -9.12 / 17.38 | 53.13 / 56.24 | 0.240 |
| | | Male / Neumark | 45 | -9.43 / 17.50 | 52.88 / 55.30 | 0.229 |
| Qwen3.6-27B | Race | Black / Bertrand | 18 | 5.26 / 8.10 | 57.96 / 60.93 | 0.062 |
| | | Black / Kline | 38 | 5.20 / 8.07 | 46.37 / 45.61 | 0.231 |
| | | White / Bertrand | 18 | 5.20 / 8.18 | 55.30 / 61.95 | 0.094 |
| | | White / Kline | 38 | 5.22 / 8.22 | 62.26 / 62.77 | 0.251 |
| | | White / Farber | 12 | 5.19 / 8.12 | 58.41 / 60.51 | 0.111 |
| | | White / Neumark | 122 | 5.18 / 8.17 | 53.69 / 55.70 | 0.168 |
| | | Female / Bertrand | 18 | 5.19 / 8.03 | 60.75 / 62.87 | 0.083 |
| | | Female / Kline | 38 | 5.20 / 8.05 | 55.49 / 52.14 | 0.241 |
| | | Female / Farber | 12 | 5.19 / 8.12 | 58.41 / 60.51 | 0.111 |
| | | Female / Neumark | 77 | 5.17 / 8.10 | 54.16 / 55.93 | 0.132 |
| | | Male / Bertrand | 18 | 5.27 / 8.25 | 52.50 / 60.01 | 0.074 |
| | | Male / Kline | 38 | 5.22 / 8.23 | 53.13 / 56.24 | 0.240 |
| | | Male / Neumark | 45 | 5.21 / 8.29 | 52.88 / 55.30 | 0.229 |
| Qwen3.6-35B-A3B | Race | Black / Bertrand | 18 | 0.17 / 0.39 | 57.96 / 60.93 | 0.062 |
| | | Black / Kline | 38 | 0.17 / 0.39 | 46.37 / 45.61 | 0.231 |
| | | White / Bertrand | 18 | 0.19 / 0.40 | 55.30 / 61.95 | 0.094 |
| | | White / Kline | 38 | 0.18 / 0.40 | 62.26 / 62.77 | 0.251 |
| | | White / Farber | 12 | 0.19 / 0.40 | 58.41 / 60.51 | 0.111 |
| | | White / Neumark | 122 | 0.18 / 0.40 | 53.69 / 55.70 | 0.168 |
| | | Female / Bertrand | 18 | 0.18 / 0.39 | 60.75 / 62.87 | 0.083 |
| | | Female / Kline | 38 | 0.18 / 0.40 | 55.49 / 52.14 | 0.241 |
| | | Female / Farber | 12 | 0.19 / 0.40 | 58.41 / 60.51 | 0.111 |
| | | Female / Neumark | 77 | 0.18 / 0.40 | 54.16 / 55.93 | 0.132 |
| | | Male / Bertrand | 18 | 0.18 / 0.40 | 52.50 / 60.01 | 0.074 |
| | | Male / Kline | 38 | 0.18 / 0.40 | 53.13 / 56.24 | 0.240 |

*Tab. S.10 continued*

| Model | Axis | Group × Study | $n$ | Model warmth/competence | Human warmth/competence | Human callback |
|---|---|---|---|---|---|---|
| | | Male / Neumark | 45 | 0.17 / 0.39 | 52.88 / 55.30 | 0.229 |

## Name-Level Warmth/Competence by Crossed Race × Gender

Tab. S.11 is the raw-value companion to Tab. 9 in the main text: race and gender crossed into four joint cells (Black-Female, Black-Male, White-Female, White-Male), recovering the interaction structure the marginal breakdown above collapses. Within each cell, callback rates are further broken down by source study rather than blended together, for the same reason as above. Cell sizes range from a single-study group of 9 names (each Black cell under Bertrand alone) to 91 distinct names, 117 name-study observations, in White-Female overall, so the crossed cell means should be read as descriptive rather than as adequately powered group comparisons.

**Table S.11: Name-level warmth/competence and callback margin by crossed race × gender group, broken down by source study, raw values.** Applicant names are joined to Gallo et al. (2024)'s published race/gender/callback labels by lowercase first name and matching study (`src/hiring_r4.py`); a name rated under more than one source study (e.g. Bertrand and Kline) contributes one row per study rather than being collapsed to a single value, since callback rates differ meaningfully by study. Model warmth/competence are raw projections and are not comparable in magnitude across models or against the 0-100 human warmth/competence scale; see Tab. 9 for the standardized ($z$-score), study-collapsed comparison.

| Model | Race × Gender | Study | $n$ | Model warmth/competence | Human warmth/competence | Human callback | Model margin |
|---|---|---|---|---|---|---|---|
| Gemma-3-12B | Black-Female | Bertrand | 9 | 37660.74 / 40372.96 | 60.95 / 61.19 | 0.065 | -0.181 |
| | | Kline | 19 | 37546.01 / 40249.57 | 46.16 / 43.04 | 0.227 | -0.184 |
| | Black-Male | Bertrand | 9 | 37842.95 / 40574.00 | 54.97 / 60.66 | 0.059 | -0.278 |
| | | Kline | 19 | 37862.93 / 40599.59 | 46.58 / 48.18 | 0.234 | -0.296 |
| | White-Female | Bertrand | 9 | 38623.69 / 41404.47 | 60.55 / 64.55 | 0.100 | -0.208 |
| | | Kline | 19 | 38636.22 / 41418.95 | 64.83 / 61.24 | 0.255 | -0.184 |
| | | Farber | 12 | 38682.20 / 41473.86 | 58.41 / 60.51 | 0.111 | -0.156 |
| | | Neumark | 77 | 38474.98 / 41250.92 | 54.16 / 55.93 | 0.132 | -0.162 |
| | White-Male | Bertrand | 9 | 38401.65 / 41170.59 | 50.04 / 59.36 | 0.089 | -0.222 |
| | | Kline | 19 | 38431.41 / 41209.65 | 59.68 / 64.30 | 0.247 | -0.250 |
| | | Neumark | 45 | 38284.64 / 41048.87 | 52.88 / 55.30 | 0.229 | -0.297 |
| Gemma-3-27B | Black-Female | Bertrand | 9 | 26107.43 / 25164.89 | 60.95 / 61.19 | 0.065 | 1.514 |
| | | Kline | 19 | 25997.99 / 25053.18 | 46.16 / 43.04 | 0.227 | 1.671 |
| | Black-Male | Bertrand | 9 | 26274.15 / 25309.69 | 54.97 / 60.66 | 0.059 | 1.778 |
| | | Kline | 19 | 26264.25 / 25303.71 | 46.58 / 48.18 | 0.234 | 1.783 |
| | White-Female | Bertrand | 9 | 26846.20 / 25851.30 | 60.55 / 64.55 | 0.100 | 1.125 |
| | | Kline | 19 | 26817.69 / 25821.04 | 64.83 / 61.24 | 0.255 | 1.059 |
| | | Farber | 12 | 26786.33 / 25790.22 | 58.41 / 60.51 | 0.111 | 0.896 |
| | | Neumark | 77 | 26626.71 / 25638.55 | 54.16 / 55.93 | 0.132 | 1.016 |
| | White-Male | Bertrand | 9 | 26662.64 / 25668.05 | 50.04 / 59.36 | 0.089 | 1.333 |
| | | Kline | 19 | 26652.83 / 25660.59 | 59.68 / 64.30 | 0.247 | 1.336 |
| | | Neumark | 45 | 26436.70 / 25453.09 | 52.88 / 55.30 | 0.229 | 1.267 |
| Llama-3.1-8B | Black-Female | Bertrand | 9 | -1.26 / -0.15 | 60.95 / 61.19 | 0.065 | -1.979 |
| | | Kline | 19 | -1.25 / -0.15 | 46.16 / 43.04 | 0.227 | -1.957 |
| | Black-Male | Bertrand | 9 | -1.27 / -0.18 | 54.97 / 60.66 | 0.059 | -2.069 |
| | | Kline | 19 | -1.27 / -0.17 | 46.58 / 48.18 | 0.234 | -2.039 |
| | White-Female | Bertrand | 9 | -1.35 / -0.17 | 60.55 / 64.55 | 0.100 | -2.021 |
| | | Kline | 19 | -1.35 / -0.17 | 64.83 / 61.24 | 0.255 | -2.033 |
| | | Farber | 12 | -1.34 / -0.17 | 58.41 / 60.51 | 0.111 | -2.062 |
| | | Neumark | 77 | -1.33 / -0.17 | 54.16 / 55.93 | 0.132 | -2.071 |
| | White-Male | Bertrand | 9 | -1.35 / -0.18 | 50.04 / 59.36 | 0.089 | -2.076 |
| | | Kline | 19 | -1.36 / -0.18 | 59.68 / 64.30 | 0.247 | -2.079 |
| | | Neumark | 45 | -1.34 / -0.17 | 52.88 / 55.30 | 0.229 | -2.107 |
| Gemma-4-12B | Black-Female | Bertrand | 9 | 12.19 / 18.40 | 60.95 / 61.19 | 0.065 | 17.121 |
| | | Kline | 19 | 12.21 / 18.41 | 46.16 / 43.04 | 0.227 | 17.127 |
| | Black-Male | Bertrand | 9 | 12.22 / 18.51 | 54.97 / 60.66 | 0.059 | 17.266 |
| | | Kline | 19 | 12.26 / 18.63 | 46.58 / 48.18 | 0.234 | 17.199 |
| | White-Female | Bertrand | 9 | 12.30 / 19.06 | 60.55 / 64.55 | 0.100 | 17.222 |
| | | Kline | 19 | 12.29 / 19.07 | 64.83 / 61.24 | 0.255 | 17.161 |
| | | Farber | 12 | 12.26 / 19.06 | 58.41 / 60.51 | 0.111 | 17.186 |
| | | Neumark | 77 | 12.25 / 18.96 | 54.16 / 55.93 | 0.132 | 17.159 |
| | White-Male | Bertrand | 9 | 12.31 / 19.06 | 50.04 / 59.36 | 0.089 | 17.073 |
| | | Kline | 19 | 12.32 / 19.10 | 59.68 / 64.30 | 0.247 | 17.128 |
| | | Neumark | 45 | 12.36 / 19.01 | 52.88 / 55.30 | 0.229 | 17.132 |
| Gemma-4-26B-A4B | Black-Female | Bertrand | 9 | 1.14 / 5.56 | 60.95 / 61.19 | 0.065 | 21.840 |
| | | Kline | 19 | 1.18 / 5.58 | 46.16 / 43.04 | 0.227 | 21.803 |
| | Black-Male | Bertrand | 9 | 0.98 / 5.49 | 54.97 / 60.66 | 0.059 | 21.160 |
| | | Kline | 19 | 0.96 / 5.54 | 46.58 / 48.18 | 0.234 | 21.158 |
| | White-Female | Bertrand | 9 | 0.92 / 5.40 | 60.55 / 64.55 | 0.100 | 21.271 |
| | | Kline | 19 | 0.92 / 5.40 | 64.83 / 61.24 | 0.255 | 21.280 |

*Tab. S.11 continued*

| Model | Race × Gender | Study | $n$ | Model warmth/competence | Human warmth/competence | Human callback | Model margin |
|---|---|---|---|---|---|---|---|
| | | Farber | 12 | 0.88 / 5.42 | 58.41 / 60.51 | 0.111 | 21.229 |
| | | Neumark | 77 | 0.92 / 5.44 | 54.16 / 55.93 | 0.132 | 21.280 |
| | White-Male | Bertrand | 9 | 0.84 / 5.48 | 50.04 / 59.36 | 0.089 | 20.840 |
| | | Kline | 19 | 0.81 / 5.45 | 59.68 / 64.30 | 0.247 | 20.793 |
| | | Neumark | 45 | 0.86 / 5.43 | 52.88 / 55.30 | 0.229 | 20.587 |
| Gemma-4-31B | Black-Female | Bertrand | 9 | 45.20 / 78.05 | 60.95 / 61.19 | 0.065 | 25.889 |
| | | Kline | 19 | 45.21 / 77.98 | 46.16 / 43.04 | 0.227 | 25.888 |
| | Black-Male | Bertrand | 9 | 45.41 / 78.30 | 54.97 / 60.66 | 0.059 | 25.781 |
| | | Kline | 19 | 45.45 / 78.44 | 46.58 / 48.18 | 0.234 | 25.789 |
| | White-Female | Bertrand | 9 | 45.56 / 79.05 | 60.55 / 64.55 | 0.100 | 25.924 |
| | | Kline | 19 | 45.51 / 78.96 | 64.83 / 61.24 | 0.255 | 25.888 |
| | | Farber | 12 | 45.48 / 78.87 | 58.41 / 60.51 | 0.111 | 25.888 |
| | | Neumark | 77 | 45.44 / 78.75 | 54.16 / 55.93 | 0.132 | 25.798 |
| | White-Male | Bertrand | 9 | 45.61 / 78.98 | 50.04 / 59.36 | 0.089 | 25.694 |
| | | Kline | 19 | 45.60 / 78.96 | 59.68 / 64.30 | 0.247 | 25.708 |
| | | Neumark | 45 | 45.58 / 78.76 | 52.88 / 55.30 | 0.229 | 25.024 |
| Qwen3-14B | Black-Female | Bertrand | 9 | -8.39 / 17.61 | 60.95 / 61.19 | 0.065 | 1.806 |
| | | Kline | 19 | -8.27 / 17.54 | 46.16 / 43.04 | 0.227 | 1.822 |
| | Black-Male | Bertrand | 9 | -8.57 / 17.11 | 54.97 / 60.66 | 0.059 | 1.667 |
| | | Kline | 19 | -8.76 / 17.07 | 46.58 / 48.18 | 0.234 | 1.599 |
| | White-Female | Bertrand | 9 | -9.11 / 17.92 | 60.55 / 64.55 | 0.100 | 1.847 |
| | | Kline | 19 | -9.13 / 17.81 | 64.83 / 61.24 | 0.255 | 1.829 |
| | | Farber | 12 | -9.21 / 17.72 | 58.41 / 60.51 | 0.111 | 1.917 |
| | | Neumark | 77 | -9.07 / 17.71 | 54.16 / 55.93 | 0.132 | 1.836 |
| | White-Male | Bertrand | 9 | -9.47 / 17.59 | 50.04 / 59.36 | 0.089 | 1.514 |
| | | Kline | 19 | -9.48 / 17.68 | 59.68 / 64.30 | 0.247 | 1.526 |
| | | Neumark | 45 | -9.43 / 17.50 | 52.88 / 55.30 | 0.229 | 1.408 |
| Qwen3.6-27B | Black-Female | Bertrand | 9 | 5.22 / 8.00 | 60.95 / 61.19 | 0.065 | 5.431 |
| | | Kline | 19 | 5.22 / 8.02 | 46.16 / 43.04 | 0.227 | 5.414 |
| | Black-Male | Bertrand | 9 | 5.30 / 8.20 | 54.97 / 60.66 | 0.059 | 5.361 |
| | | Kline | 19 | 5.18 / 8.11 | 46.58 / 48.18 | 0.234 | 5.362 |
| | White-Female | Bertrand | 9 | 5.16 / 8.06 | 60.55 / 64.55 | 0.100 | 5.403 |
| | | Kline | 19 | 5.18 / 8.09 | 64.83 / 61.24 | 0.255 | 5.414 |
| | | Farber | 12 | 5.19 / 8.12 | 58.41 / 60.51 | 0.111 | 5.458 |
| | | Neumark | 77 | 5.17 / 8.10 | 54.16 / 55.93 | 0.132 | 5.365 |
| | White-Male | Bertrand | 9 | 5.23 / 8.30 | 50.04 / 59.36 | 0.089 | 5.264 |
| | | Kline | 19 | 5.27 / 8.34 | 59.68 / 64.30 | 0.247 | 5.296 |
| | | Neumark | 45 | 5.21 / 8.29 | 52.88 / 55.30 | 0.229 | 5.189 |
| Qwen3.6-35B-A3B | Black-Female | Bertrand | 9 | 0.17 / 0.39 | 60.95 / 61.19 | 0.065 | 3.167 |
| | | Kline | 19 | 0.17 / 0.39 | 46.16 / 43.04 | 0.227 | 3.086 |
| | Black-Male | Bertrand | 9 | 0.17 / 0.39 | 54.97 / 60.66 | 0.059 | 2.903 |
| | | Kline | 19 | 0.17 / 0.39 | 46.58 / 48.18 | 0.234 | 2.862 |
| | White-Female | Bertrand | 9 | 0.19 / 0.40 | 60.55 / 64.55 | 0.100 | 3.250 |
| | | Kline | 19 | 0.19 / 0.40 | 64.83 / 61.24 | 0.255 | 3.237 |
| | | Farber | 12 | 0.19 / 0.40 | 58.41 / 60.51 | 0.111 | 3.229 |
| | | Neumark | 77 | 0.18 / 0.40 | 54.16 / 55.93 | 0.132 | 3.170 |
| | White-Male | Bertrand | 9 | 0.18 / 0.41 | 50.04 / 59.36 | 0.089 | 2.847 |
| | | Kline | 19 | 0.18 / 0.40 | 59.68 / 64.30 | 0.247 | 2.862 |
| | | Neumark | 45 | 0.17 / 0.39 | 52.88 / 55.30 | 0.229 | 2.808 |

# Pending Updates (Internal Tracking, Remove Before Submission)

The nine-model local hiring-steering sweep (Step 31, 2026-08-05) closed the raw/denoised-local coverage gap for Llama-3.1-8B and Qwen3-14B, but the following Results-facing artifacts still use their old broad-regime ($\alpha = +0.50$) data for these two models and need a follow-up pass:

- **Transition summary table**: `ModelSpec` entries for Llama-3.1-8B and Qwen3-14B in `paper/figures/_steering_transition_flow_common.py` still point at the broad CSVs (`hiring_steering_raw_llama31_8b.csv` and `hiring_steering_raw_qwen3_14b.csv`) at $\alpha = 0.50$; update to the new `..._local.csv` files at $\alpha = 0.10$, matching the other seven models, then regenerate `results/tables/hiring_steering_transition_summary_9model.csv / .tex` and `paper_figure4_hiring_bidirectional_examples.pdf`.
- **Stale caveat sentence**: the paragraph before `\input` of `hiring_steering_transition_summary_9model.tex` states that "Llama-3.1-8B and Qwen3-14B use their available broad-regime endpoint of $\alpha = +0.50$, so raw effect sizes should not be compared directly across all rows." Remove or rewrite once the table above is regenerated on local-regime data.
- **Transition-direction claims**: the sentence "Positive steering crosses the mean decision boundary in both directions among the Gemma-3 models and from No to Yes for Llama-3.1-8B" was written against broad-regime data for Llama-3.1-8B; re-check it against the regenerated local-regime table, since the transition pattern may differ at $\alpha = 0.10$.
- **New candidate finding**: Step 31 found Qwen3-14B's competence steering slope flips sign between the broad regime (negative, $\Delta_{+0.5} - \Delta_{-0.5} = -0.285$) and the local regime (positive, $\Delta_{+0.1} - \Delta_{-0.1} = +0.204$), the same qualitative pattern that motivated the original Gemma-3-27B local sweep. Decide whether this belongs in the Results narrative once written.
- `supp_figure1_hiring_competence_transitions.pdf` exists as a generated artifact but is not currently referenced anywhere in this manuscript; decide whether it should be included or left unused.
- **Denoised audit/disparity/mediation not run**: `src/hiring_audit.py` has no `--vector-kind` option and has only ever been run on raw vectors, so probe-vs-human correlation, group disparity, and mediation have no denoised robustness check (unlike concept steering and hiring steering). Consider adding the flag and re-running for all nine models, mirroring the denoised local hiring-steering sweep added in Step 31.
- **Mediation prose references a parked Discussion argument**: the Results mediation paragraph frames the Gemma-3 null as a steerability-paradox observation; the fuller version of that argument currently sits in the parked Discussion. Re-check and realign this cross-reference when the Discussion is rebuilt.